

User Acceptance Testing (UAT) Document

CampusSafe Twin Campus Safety & Incident Management System

NWU Potchefstroom Campus • Digital Twin Safety Platform

TEAM

Student number	Surname and initials	UAT role
49019767	Hattingh P	UAT Coordinator
53324757	Netshiukhwi MC	Tester: Student module (lead)
46105948	Mazibuko SN	Tester: Student module
50763458	Mlambo M	Tester: Security officer module (lead)
45698260	Bangane I	Tester: Security officer module
47742488	Mdluli D	Tester: Supervisor module
50881434	Malima M	Tester: Supervisor module
46330933	Maseko S	Re-verification & document QA

Document version 1.1

UAT round	Round 1 (Full regression of all modules)
Execution date	05 October 2026
Compiled by	Hattingh P (UAT Coordinator), with the UAT team listed above
Execution method	Scripted end-to-end browser walkthrough (Microsoft Edge)
Overall result	25 / 25 test cases passed , 4 minor observations logged
Remediation	All 4 observations fixed and re-verified on 05 October 2026 (Section 7.1)
Status	Executed & remediated, awaiting business sign off (Section 8)

This document contains the UAT plan, test environment description, 25 detailed test cases with step by step instructions, expected and actual results, 31 captured screen evidence figures, an observation log, the UAT team work allocation, and the formal acceptance sign off sheet.

1. Document Control

1.1 Version History

Version	Date	Author	Description
1.0	2026-10-05	Hattingh P	Initial UAT document: Round 1 execution results for all system modules, with full screen evidence.
1.1	2026-10-05	Hattingh P	Remediation update: all four Round 1 observations (OBS-01 to OBS-04) fixed, re-verified in the live environment and evidenced in new Section 7.1 (Figures 28–31). Acceptance statement updated accordingly.

1.2 UAT Team & Work Allocation

Student no.	Name	UAT role	Work carried out
49019767	Hattingh P	UAT Coordinator	• UAT planning: strategy, scope, entry/exit and pass/fail criteria (Section 2, Section 3) • Designed and scripted all 25 test cases and allocated them to the testers • Set up the test environment, demo accounts and seeded test data (Section 4) • Executed Module A (UAT-A1 to UAT-A6) and the cross-role SOS chain (UAT-B1, UAT-B2) • Kept the observation log, triaged OBS-01 to OBS-04 with the developers and coordinated re-verification (Section 7) • Consolidated all evidence, compiled and edited this document
53324757	Netshiukhwi MC	Tester: Student module (lead)	• Executed UAT-B3, UAT-B4 and UAT-B5 (incident reporting, My Reports, report timeline) • Captured Figure 8 to Figure 11 and checked the student module evidence for completeness
46105948	Mazibuko SN	Tester: Student module	• Executed UAT-B6, UAT-B7 and UAT-B8 (safe route planner, notifications, Help Centre) • Captured Figure 12 to Figure 14
50763458	Mlambo M	Tester: Security officer module (lead)	• Executed UAT-C1, UAT-C2 and UAT-C3 (security map, incident detail, dispatch queue) • Captured Figure 16 to Figure 18; logged OBS-02 and OBS-03
45698260	Bangane I	Tester: Security officer module	• Executed UAT-C4 and UAT-C5 (patrol hotspots, flagged safety zones); captured Figure 19 and Figure 20 • Verified the test data baseline before and after execution (Section 4.3)
47742488	Mdluli D	Tester: Supervisor module	• Executed UAT-D1, UAT-D2 and UAT-D3 (Live Ops, route network editor, campus locations) • Captured Figure 21 to Figure 23
50881434	Malima M	Tester: Supervisor module	• Executed UAT-D4, UAT-D5 and UAT-D6 (staff administration, analytics, simulation) • Captured Figure 24 to Figure 26; logged OBS-04
46330933	Maseko S	Re-verification & document QA	• Re verified the fixes for OBS-01 to OBS-04 in the live UAT environment; captured Figure 28 to Figure 31 (Section 7.1) • Independent review and proofreading of this document; prepared the sign-off sheet (Section 8)

2. Introduction

2.1 Purpose

This document records the User Acceptance Testing of the **CampusSafe Twin** campus safety portal. Its purpose is to demonstrate, with captured screen evidence, that the system satisfies the business requirements for each of its three user roles, and to provide the formal basis on which the business stakeholders accept (or reject) the system for operational use.

2.2 System Overview

CampusSafe Twin is a web based (PWA) campus safety and incident management platform ("digital twin") for the NWU Potchefstroom campus. It supports three user roles with distinct capabilities:

Role	Capabilities
Student	One-tap emergency SOS with automatic guard dispatch; incident reporting with map pin and photo evidence; report status tracking with timeline; safer walking route planner; notifications; help centre.
Security Officer (Guard)	Live campus incident map with open incident queue; dispatch queue with accept / decline workflow; incident detail management (notes, status lifecycle, evidence); patrol hotspot intelligence; flagged safety zones.
Supervisor	All guard capabilities plus: live operations command view; walkway routing network editor; campus locations management; staff account administration; safety analytics with CSV export; simulation / demo data generator.

Technology stack: ASP.NET WebForms (.NET Framework 4.8) on IIS Express, PostgreSQL 16.9 with PostGIS 3.6.2 (spatial queries, routing network), Npgsql 4.1.14 data access, Leaflet maps with Esri satellite imagery, OSM derived walkway routing graph (2,576 nodes / 2,777 links) with Dijkstra based safe route computation.

2.3 Scope

In scope (this UAT round)

- Authentication, student self registration, role-based navigation and URL access control (Module A)
- Student workflows: SOS, incident reporting, report tracking, safe route, notifications, help (Module B)
- Security officer workflows: live map, incident queue and detail, dispatches, hotspots, flagged zones (Module C)
- Supervisor workflows: live ops, route network, locations, staff admin, analytics, simulation (Module D)

Out of scope

- Performance / load / stress testing and penetration testing
- External integrations (e mail / SMS gateways): none are configured in this release
- Native mobile applications (system is responsive web only)
- Backup / restore and disaster recovery procedures

3. UAT Approach

3.1 Strategy

Testing was executed as **role based end to end scenarios** in a real Chromium browser (Microsoft Edge) against a fully provisioned local environment with representative seeded data. Where a workflow creates data (SOS alert, incident report, registration), the created record was traced through to the other roles' screens to verify cross-role integration (e.g. a student's SOS appearing in the guard's dispatch queue). Every test case includes at least one captured screenshot as acceptance evidence; figures are referenced from each test case and reproduced in full in [Section 6.1](#).

3.2 Entry Criteria (all met)

- Application builds and deploys successfully; environment starts via `start_app.ps1`
- Database schema installed; seeded demo dataset present (users, 60 locations, walkway network, 40+ incidents)
- Demo accounts available for each role ([Section 4.2](#))

3.3 Exit Criteria

- 100% of planned test cases executed
- No open Severity 1 (Critical) or Severity 2 (Major) defects
- All Severity 3/4 observations logged with a triage recommendation ([Section 7](#))
- Business sign-off obtained ([Section 8](#))

3.4 Pass / Fail Criteria

Verdict	Definition
PASS	Actual behaviour matches the expected result; any deviation is cosmetic and logged as an observation.
FAIL	Actual behaviour deviates from the expected result in a way that prevents or corrupts the business workflow.
BLOCKED	Test could not be executed due to an environmental issue or a failed prerequisite test.

3.5 Defect Severity Classification

Severity	Definition
Sev 1 – Critical	Safety critical function unusable (e.g. SOS fails to dispatch); data loss; security breach.
Sev 2 – Major	A core workflow fails with no workaround.
Sev 3 – Minor	Workflow completes but with incorrect or confusing behaviour; workaround exists.
Sev 4 – Cosmetic	Visual, text or layout issue with no functional impact.

4. Test Environment & Data

4.1 Environment

Component	Detail
Application URL	http://localhost:52000 (IIS Express)
Application platform	ASP.NET WebForms, .NET Framework 4.8
Database	PostgreSQL 16.9 + PostGIS 3.6.2, database <code>campussafe</code> , Hosted on SupaBase
Data access	Npgsql 4.1.14
Browser used for UAT	Microsoft Edge (Chromium), 1400x900 viewport, geolocation granted at campus centre (-26.6890, 27.0935)
Mapping	Leaflet + Esri satellite imagery; OSM-derived walkway network
Host	Windows 11 Pro (local UAT environment)

4.2 Test Accounts

Role	Account	Origin
Student	<code>sipho@student.nwu.ac.za</code> (Sipho Dlamini)	Seeded demo account
Security Officer	<code>kobus@guard.nwu.ac.za</code> (Officer Kobus van der Merwe)	Seeded demo account
Supervisor	<code>pieta@security.nwu.ac.za</code> (Pieta Supervisor)	Seeded demo account
Student (new)	<code>thandi.uat@student.nwu.ac.za</code> (Thandi Mokoena)	Created during UAT via self-registration (UAT-A3)

All demo accounts use the shared UAT password distributed with the environment start script. Credentials are for the isolated local UAT environment only.

4.3 Test Data Summary (state at start of execution)

Dataset	Volume
Campus locations / POIs	60 named locations (OSM-derived)
Walkway routing network	2,576 nodes / 2,777 links (largest connected component of walkable OSM ways)
Logged incidents (incl. simulation batches)	44 at start; 46 after UAT (1 SOS + 1 report created during execution)
User accounts	4 at start; 5 after UAT (1 registered during execution)
Notifications	10+

5. Test Execution Summary

25 PLANNED	25 EXECUTED	25 PASSED	0 FAILED	4 OBSERVATIONS
----------------------	-----------------------	---------------------	--------------------	--------------------------

Click a test case ID to jump to its detailed table in [Section 6](#), or a figure number to jump to that screenshot in [Section 6.1](#).

ID	Test case	Role	Tester	Result	Evidence
UAT-A1	Login page renders with branding and credential fields	Public	Hattingh P	PASS	Fig. 1
UAT-A2	Invalid credentials are rejected with a clear error	Public	Hattingh P	PASS	Fig. 2
UAT-A3	New student can self-register and is directed to sign in	Public	Hattingh P	PASS	Figs. 3 , 4 , 5
UAT-A4	Valid login lands each role on its correct start page	All	Hattingh P	PASS	Figs. 6 , 16 , 19
UAT-A5	URL access control blocks unauthorised pages per role	Student	Hattingh P	PASS	Fig. 15
UAT-A6	Logout terminates the session and returns to login	All	Hattingh P	PASS	Fig. 27
UAT-B1	Emergency SOS triggers with geolocation and reference number	Student	Hattingh P	PASS	Figs. 6 , 7
UAT-B2	SOS auto-dispatches the nearest guard and alerts security screens	Student → Guard	Hattingh P	PASS	Figs. 16 , 18
UAT-B3	Incident report with category, date/time, map pin and photo	Student	Netshiukhwi MC	PASS	Figs. 8 , 9
UAT-B4	Submitted reports listed under My Reports with status badges	Student	Netshiukhwi MC	PASS	Fig. 10
UAT-B5	Report timeline shows status history for a selected report	Student	Netshiukhwi MC	PASS	Fig. 11
UAT-B6	Safe route planner computes safer vs shortest route with metrics	Student	Mazibuko SN	PASS	Fig. 12
UAT-B7	Notifications page lists messages with unread counter in nav	Student	Mazibuko SN	PASS	Fig. 13
UAT-B8	Help Centre is reachable from every page and searchable	Student	Mazibuko SN	PASS	Fig. 14
UAT-C1	Live security map with status legend and open incidents queue	Guard	Mlambo M	PASS	Fig. 16
UAT-C2	Incident detail: information, evidence, notes, lifecycle action	Guard	Mlambo M	PASS	Fig. 17
UAT-C3	Dispatch queue with accept / decline and tactical map	Guard	Mlambo M	PASS	Fig. 18
UAT-C4	Patrol hotspot intelligence with top patrol areas	Guard	Bangane I	PASS	Fig. 19
UAT-C5	Flagged safety zones: view and create	Guard	Bangane I	PASS	Fig. 20
UAT-D1	Live Operations command view	Supervisor	Mdluli D	PASS	Fig. 21
UAT-D2	Walkway routing network editor loads the OSM network	Supervisor	Mdluli D	PASS	Fig. 22

ID	Test case	Role	Tester	Result	Evidence
UAT-D3	Campus locations management (list, map picker, add, seed)	Supervisor	Mdluli D	PASS	Fig. 23
UAT-D4	Staff account administration (list, create account)	Supervisor	Malima M	PASS	Fig. 24
UAT-D5	Analytics dashboard: KPIs, distributions, dispatch history, CSV export	Supervisor	Malima M	PASS	Fig. 25
UAT-D6	Simulation / demo data generator with purge controls	Supervisor	Malima M	PASS	Fig. 26

5.1 Observation Summary

ID	Observation	Severity	Evidence	Status
OBS-01	SOS progress tracker step 2 icon renders as garbled characters ("âœ") instead of a check mark (character-encoding issue).	Sev 4	Fig. 7	FIXED
OBS-02	"Action" buttons in the Security Map open-incidents queue render without a visible label.	Sev 4	Fig. 16	FIXED
OBS-03	Tactical map on My Dispatches shows "Map data not yet available" tiles at its default zoom level.	Sev 3	Fig. 18	FIXED
OBS-04	Analytics shows "N/A" for average response / resolution time while dispatch history rows exist (arrivals pending).	Sev 4	Fig. 25	FIXED

Details and recommendations in [Section 7](#); remediation and reverification evidence in [Section 7.1](#). None of the observations blocked a business workflow; all 25 test cases therefore carry a PASS verdict, and all four observations were subsequently fixed and re-verified on the same day.

6. Detailed Test Cases & Evidence

Module A – Authentication, Registration & Access Control

UAT-A1 – Login page renders correctly		PASS
Role / user	Public (not signed in)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Application running at <code>http://localhost:52000</code>	
Steps	1. Open <code>/Login.aspx</code> in the browser.	
Expected	Branded login page with email and password fields, Sign In button, a link to student registration, and Login / Register links in the header.	
Actual	As expected: CampusSafe branded card with both credential fields, Sign In button and "Create student account" link displayed.	
Evidence	Figure 1	

UAT-A2 – Invalid credentials rejected		PASS
Role / user	Public	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	On the login page	
Steps	1. Enter a valid account email (<code>sipho@student.nwu.ac.za</code>) with an incorrect password. 2. Click Sign In .	
Expected	Login refused; a clear error message shown; no session created; password not retained.	
Actual	As expected: user remains on the login page with the banner "Invalid email or password."; the password field is cleared.	
Evidence	Figure 2	

UAT-A3 – Student self-registration		PASS
Role / user	Public → new student	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Email <code>thandi.uat@student.nwu.ac.za</code> not yet registered	
Steps	1. Open <code>/Register.aspx</code> . 2. Complete full name, email, student number, password and password confirmation. 3. Click Create Student Account .	
Expected	Account created; user redirected to the login page with a success confirmation; new account can be used to sign in.	
Actual	As expected: redirected to <code>/Login.aspx?registered=1</code> showing "Registration successful! You may now sign in with your email and password."	
Evidence	Figures 3 , 4 , 5	

UAT-A4 – Role-based landing & navigation		PASS
Role / user	Student, Guard, Supervisor (all three demo accounts)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Demo accounts active	
Steps	1. Sign in as each role in turn. 2. Observe the landing page and the navigation menu offered.	
Expected	Student lands on incident reporting with the student menu (SOS, Report Incident, My Reports, Safe Route, Notifications). Guard lands on the Security Map with the guard menu. Supervisor lands on Hotspots with the full supervisor menu (12 items incl. Staff, Analytics, Simulation).	
Actual	As expected for all three roles; the header greets the signed in user by name and the menu contents match the role exactly.	
Evidence	Figures 6 (student), 16 (guard), 19 (supervisor menus visible)	

UAT-A5 – URL access control enforcement		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Signed in as student	
Steps	1. Type the supervisor-only URL <code>/Admin/Users.aspx</code> directly into the address bar.	
Expected	The page is not shown; the student is redirected away from the administrative page.	
Actual	As expected: the request was redirected to the student's own start page (<code>/Student/Report.aspx</code>); no staff data was exposed.	
Evidence	Figure 15	

UAT-A6 – Logout		PASS
Role / user	Supervisor (last signed-in role)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Signed in	
Steps	1. Click Logout in the header. 2. Attempt to use the browser Back button / revisit a protected page.	
Expected	Session terminated; user returned to the login page; protected pages redirect to login when revisited.	
Actual	As expected: returned to the login screen in the signed-out state (Login / Register links shown in header).	
Evidence	Figure 27	

Module B – Student Workflows

UAT-B1 – Emergency SOS trigger		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	Signed in as student; browser geolocation granted (campus centre)	
Steps	1. Open SOS from the navigation (prominent red menu item). 2. Press the panic button. 3. Observe the SOS status panel.	
Expected	An emergency alert is created at the student's location with a unique reference; a live 4-stage status tracker is shown (Signal Sent → Officer Dispatched → En Route → On Scene); the alert can be cancelled if accidental.	
Actual	As expected: alert CS-2F214C1E created instantly; panel shows "SOS Active: Help is on the way!" with stages 1–2 already complete ("Emergency security officer notified and assigned"), an auto refreshing status and a Cancel Alert (Accidental) option. Observation OBS-01: stage-2 icon glyph garbled (cosmetic).	
Evidence	Figures 6 , 7	

UAT-B2 – SOS reaches security in real time		PASS
Role / user	Student action verified on Guard screens (Officer Kobus van der Merwe)	
Executed by	Hattingh P (UAT Coordinator)	
Preconditions	SOS CS-2F214C1E active (UAT-B1)	
Steps	1. Sign in as the security officer. 2. Open the Security Map and My Dispatches pages.	
Expected	The SOS appears to the officer as a critical alert and as an assigned dispatch that can be accepted or declined.	
Actual	As expected: the Security Map shows the red banner "EMERGENCY ALERT – CRITICAL SOS DISPATCHED TO YOU – SOS Alert near (-26.6890, 27.0935)" with Focus on Map and Open Dispatch Queue shortcuts; My Dispatches lists CS-2F214C1E • medical • EMERGENCY SOS , dispatch status "assigned", with Accept & En Route / Decline / Open Incident Details actions.	
Evidence	Figures 16 , 18	

UAT-B3 – Submit incident report (pin + photo)		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Netshiukhwi MC	
Preconditions	Signed in as student	
Steps	1. Open Report Incident. 2. Select category "Suspicious Activity"; set date 2026-10-05 and time 19:30. 3. Enter an incident description. 4. Click the campus map to drop the incident location pin. 5. Attach a JPG photo as evidence. 6. Click Submit Incident Report.	
Expected	Report accepted with a unique reference number; pinned coordinates stored; photo attached; report immediately visible to security staff.	
Actual	As expected: report CS-85BA6578 created with confirmation message; the report subsequently appears at the top of the guard's Open Incidents Queue (category suspicious_activity, occurred 2026-10-05 19:30, reporter Sipho Dlamini), see Figure 16 .	
Evidence	Figures 8 , 9	

UAT-B4 – My Reports list		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Netshiukhwi MC	
Preconditions	Student has at least one submitted report (UAT-B3)	
Steps	1. Open My Reports.	
Expected	All of the student's own reports listed with reference, occurrence date, category and a colour-coded status badge; a timeline action per report.	
Actual	As expected: grid lists the student's reports including the newly submitted CS-85BA6578 with status "New" and a "View Timeline" action.	
Evidence	Figure 10	

UAT-B5 – Report status timeline		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Netshiukhwi MC	
Preconditions	On My Reports with at least one report	
Steps	1. Click View Timeline on a report row.	
Expected	A timeline panel opens showing the report's reference, category, description and its chronological status history.	
Actual	As expected: the "Report Timeline" panel opens for the selected reference with its details and status history entries.	
Evidence	Figure 11	

UAT-B6 – Safe route planner		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Mazibuko SN	
Preconditions	Routing network and campus locations seeded	
Steps	1. Open Safe Route. 2. Select a start location ("B6") and destination ("E6 JS van der Merwe-gebou") from the dropdowns. 3. Click Find Safest Route.	
Expected	A route is computed along real walkways and drawn on the satellite map with A/B markers; comparison cards show distance, estimated walking time and risk for the recommended safer route vs the shortest route; the route can be saved; legend explains flagged zones and hotspot overlays.	
Actual	As expected: "Route calculated successfully!" with safer route 623.8 m / 8 min, risk score 0 (identical to the shortest route for this pair, as no risk factors lie on the path); purple route polyline drawn between A and B markers; flagged-zone circles and hotspot cells overlaid; Save Route panel available.	
Evidence	Figure 12	

UAT-B7 – Notifications		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Mazibuko SN	
Preconditions	Signed in; system has generated notifications (e.g. from SOS / report activity)	
Steps	1. Open Notifications from the navigation.	
Expected	Personal notifications listed newest first; unread count shown as a badge in the navigation; "Mark All as Read" available.	
Actual	As expected: "My Notifications" lists the user's messages with read/unread state and the Mark All as Read control.	
Evidence	Figure 13	

UAT-B8 – Help Centre		PASS
Role / user	Student (Sipho Dlamini)	
Executed by	Mazibuko SN	
Preconditions	Signed in (any role)	
Steps	1. Click the ? Help button in the page header. 2. Review the available guides and the search control.	
Expected	Help Centre opens with searchable how-to guides on using the system.	
Actual	As expected: Help Centre page with guide content and Search / Show All controls.	
Evidence	Figure 14	

Module C – Security Officer Workflows

UAT-C1 – Live security map & incident queue		PASS
Role / user	Guard (Officer Kobus van der Merwe)	
Executed by	Mlambo M	
Preconditions	Signed in as guard; seeded incidents present	
Steps	1. Open Security Map. 2. Review the map markers, the status legend and the Open Incidents Queue table.	
Expected	Spatial view of the last 30 days of incidents, colour-coded by status (New / Acknowledged / In Progress / Resolved); a queue table of open incidents with reference, category, status, timestamps, reporter and an action per row; duty-status control.	
Actual	As expected: satellite map rendered with dozens of colour-coded incident markers inside the campus boundary; the queue lists 30+ open incidents including those created during this UAT. Observation OBS-02: per-row action buttons lack a visible text label (cosmetic).	
Evidence	Figure 16	

UAT-C2 – Incident detail management		PASS
Role / user	Guard (Officer Kobus van der Merwe)	
Executed by	Mlambo M	
Preconditions	An open incident exists (CS-21C58B04, status "In Progress")	
Steps	1. Open the incident's detail page from the queue. 2. Review incident information, evidence section, notes and lifecycle controls.	
Expected	Detail view shows status, category, occurrence/report timestamps, reporter, coordinates, description, a location pin map, photo evidence and staff notes, an investigation-note control, and a lifecycle action to advance the status (with optional note).	
Actual	As expected for CS-21C58B04: full incident information panel; "In Progress" status badge; location pin on the mini-map; Add Note control; lifecycle action correctly offers the next status ("Mark Resolved").	
Evidence	Figure 17	

UAT-C3 – Officer dispatch queue		PASS
Role / user	Guard (Officer Kobus van der Merwe)	
Executed by	Mlambo M	
Preconditions	A dispatch is assigned to the officer (SOS from UAT-B1)	
Steps	1. Open My Dispatches.	
Expected	Assigned callouts listed with urgency, reference, coordinates, dispatch status and response actions; a tactical map distinguishes Emergency SOS, standard callouts and the officer's own patrol pin.	
Actual	As expected: dispatch card for CS-2F214C1E (EMERGENCY SOS, status "assigned") with View on Map , Accept & En Route , Decline and Open Incident Details actions; patrol pin shown on the tactical map. Observation OBS-03: satellite tiles unavailable at the tactical map's default zoom (map framework and markers otherwise functional).	
Evidence	Figure 18	

UAT-C4 – Patrol hotspot intelligence		PASS
Role / user	Guard (Officer Kobus van der Merwe)	
Executed by	Bangane I	
Preconditions	Historical incident data present	
Steps	1. Open Patrol Hotspots .	
Expected	Heat/grid visualisation of incident concentration across campus with a ranked "Top Areas to Patrol" list.	
Actual	As expected: hotspot cells rendered over the campus satellite map with the ranked patrol-priority list beneath.	
Evidence	Figure 19	

UAT-C5 – Flagged safety zones		PASS
Role / user	Guard (Officer Kobus van der Merwe)	
Executed by	Bangane I	
Preconditions	Signed in as guard or supervisor	
Steps	1. Open Flagged Zones . 2. Review existing zones and the zone-creation form (name, radius, reason, map placement).	
Expected	Existing flagged zones listed and drawn on the map; staff can define a new zone which will influence safe-route risk weighting.	
Actual	As expected: flagged zones displayed on the map with the "Create Flagged Zone" form available. Active zones were also confirmed rendering as red circles on the student Safe Route map (Figure 12).	
Evidence	Figure 20	

Module D – Supervisor Workflows

UAT-D1 – Live Operations command		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Mdluli D	
Preconditions	Signed in as supervisor; active incidents and dispatches present	
Steps	1. Open Live Ops .	
Expected	Manager command view combining live incident/dispatch state and officer positions for operational oversight.	
Actual	As expected: "Manager Live Operations Command" view rendered with the live operational map and current activity.	
Evidence	Figure 21	

UAT-D2 – Walkway routing network editor		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Mdluli D	
Preconditions	OSM network seeded (2,576 nodes / 2,777 links)	
Steps	1. Open Route Network . 2. Review the rendered walkway graph and editor controls.	
Expected	The full walkway network renders over the campus map; the supervisor has tools to maintain it (including network rebuild from source data).	
Actual	As expected: "Walkway Routing Network Editor" shows the complete OSM-derived network geometry over the satellite base map with editor controls.	
Evidence	Figure 22	

UAT-D3 – Campus locations management		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Mdluli D	
Preconditions	60 campus locations seeded	
Steps	1. Open Locations . 2. Review the location list, map picker and the add-location form.	
Expected	All campus POIs listed and mapped; supervisor can add a location via the map picker and re-seed standard POIs.	
Actual	As expected: "Campus Locations & Landmarks" lists the seeded POIs with the map picker and Save Campus Location / Seed Standard Campus POIs controls. These locations correctly populate the Safe Route endpoint dropdowns (Figure 12).	
Evidence	Figure 23	

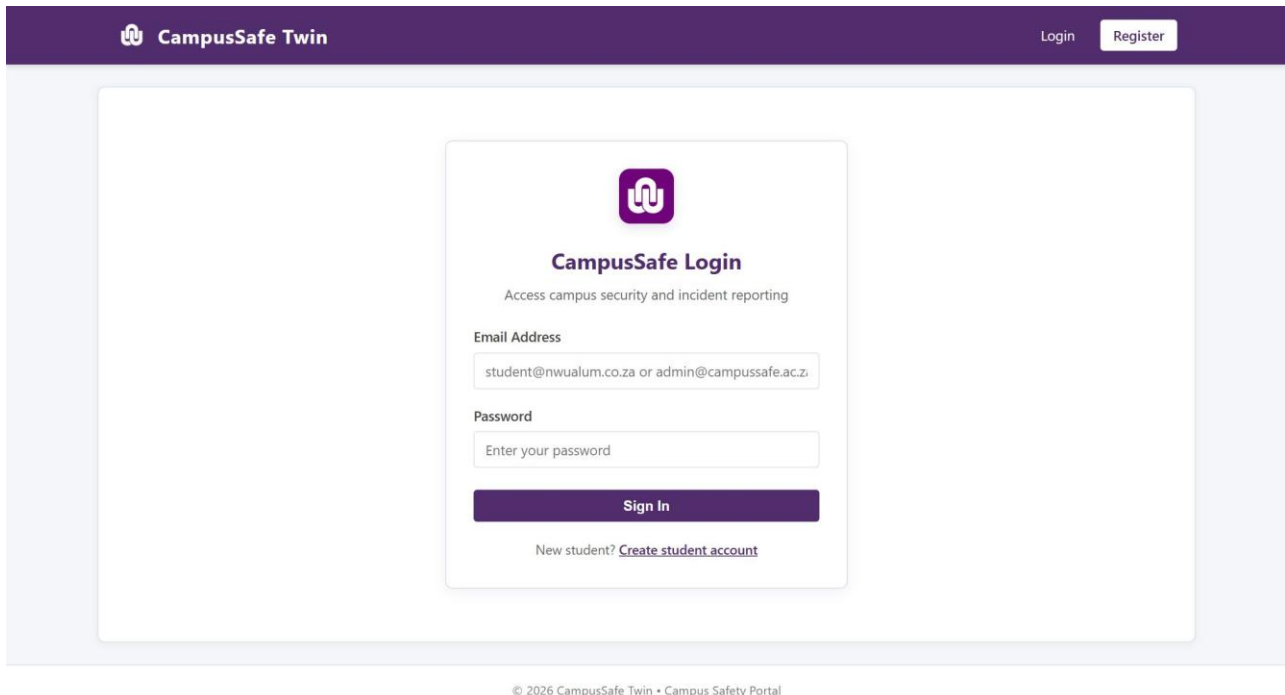
UAT-D4 – Staff account administration		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Malima M	
Preconditions	Signed in as supervisor	
Steps	1. Open Staff. 2. Review the account list and the create-account form.	
Expected	User accounts listed with role; supervisor can create guard/staff accounts. Page is inaccessible to other roles (verified in UAT-A5).	
Actual	As expected: "User Accounts & Staff Access" shows the account list (including the student registered in UAT-A3) and the Create Account form.	
Evidence	Figure 24	

UAT-D5 – Safety analytics & export		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Malima M	
Preconditions	Incident history present (46 incidents at time of test)	
Steps	1. Open Analytics. 2. Review KPI tiles, category and status distributions, and officer dispatch history. 3. Confirm the CSV export control is available.	
Expected	Dashboard shows total incidents, SOS alert count, response-time KPIs, incident category distribution, resolution status distribution and a dispatch history table; "Export All Incidents (CSV)" available.	
Actual	As expected: KPIs show 46 total incidents and 14 emergency SOS alerts (including those raised during this UAT); category distribution (medical 22%, harassment 17%, suspicious activity 15%, ...) and resolution distribution (33% resolved) render correctly; dispatch history lists the UAT SOS assigned to Officer Kobus van der Merwe. Observation OBS-04: average response/resolution KPIs display "N/A" while arrivals are pending.	
Evidence	Figure 25	

UAT-D6 – Simulation / demo data generator		PASS
Role / user	Supervisor (Pieta Supervisor)	
Executed by	Malima M	
Preconditions	Signed in as supervisor	
Steps	1. Open Simulation. 2. Review the generation parameters and batch management (purge) controls.	
Expected	Supervisor can generate batches of simulated incidents for demonstration/training and delete seeded batches; destructive purge requires confirmation.	
Actual	As expected: "Simulation & Demo Data Generator" with Generate Simulation Data form and "Delete All Demo Batches" guarded by a confirmation prompt. (Generation/purge was not executed in this round to preserve the UAT dataset.)	
Evidence	Figure 26	

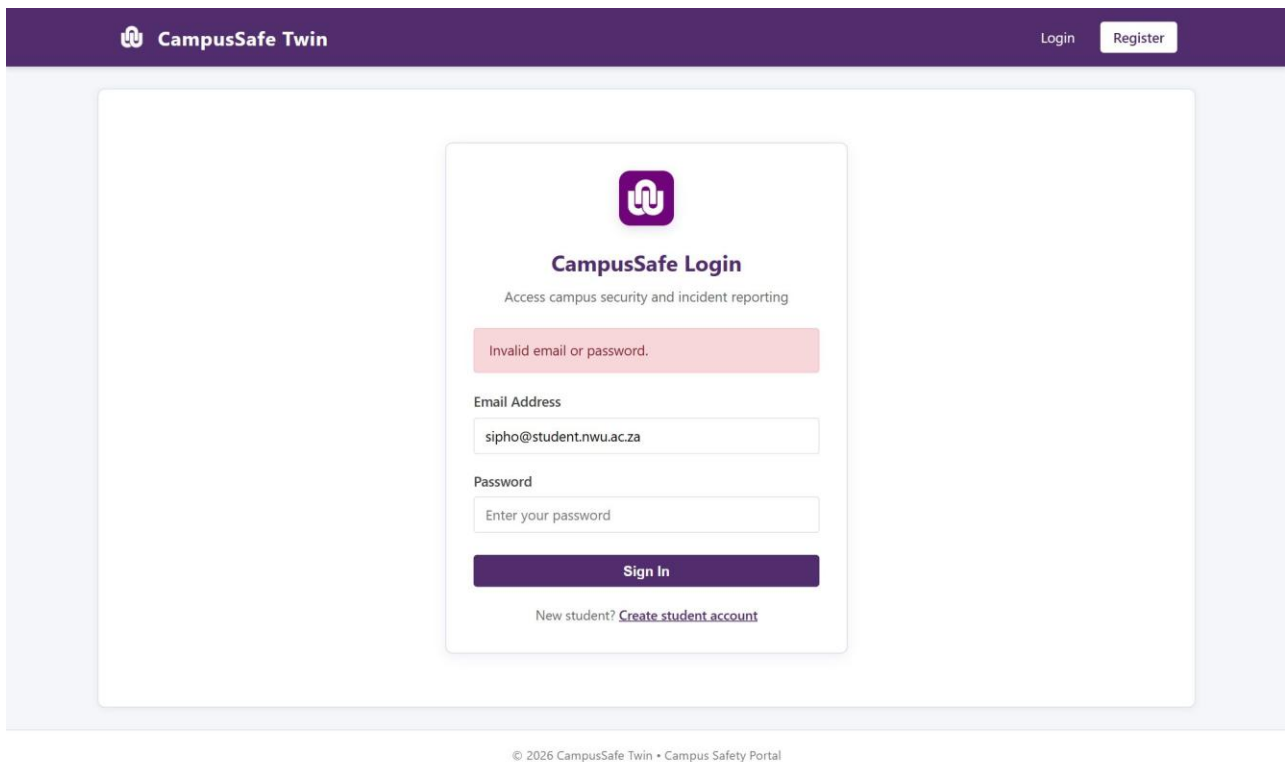
6.1 Evidence Figures

All screenshots captured on 05 October 2026 against the environment described in [Section 4](#), in execution order. The test case reference in each caption links back to the test case.



The screenshot shows the CampusSafe Login page. At the top, there is a purple header with the CampusSafe Twin logo on the left and 'Login' and 'Register' buttons on the right. The main content area is white and contains a central login form. The form has the CampusSafe logo at the top, followed by the title 'CampusSafe Login' and the subtitle 'Access campus security and incident reporting'. Below this, there are two input fields: 'Email Address' with the placeholder text 'student@nwualum.co.za or admin@campussafe.ac.za' and 'Password' with the placeholder text 'Enter your password'. A purple 'Sign In' button is positioned below the password field. At the bottom of the form, there is a link that says 'New student? [Create student account](#)'. The footer of the page is a light purple bar with the text '© 2026 CampusSafe Twin • Campus Safety Portal'.

Figure 1 – Login page ([UAT-A1](#)).



This screenshot shows the same CampusSafe Login page as Figure 1, but with an error message. A pink rectangular box with the text 'Invalid email or password.' is displayed above the 'Email Address' input field. The 'Email Address' field now contains the text 'sipho@student.nwu.ac.za'. The 'Password' field still has the placeholder 'Enter your password'. The 'Sign In' button and the 'Create student account' link remain visible. The header and footer are identical to the previous screenshot.

Figure 2 – Invalid credentials rejected with "Invalid email or password." ([UAT-A2](#)).

CampusSafe Twin

LoginRegister



Student Registration

Join CampusSafe Twin to report incidents and access safe walking routes

Full Name

University / Student Email

Student Number

Password

Confirm Password

Create Student Account

Already have an account? [Sign in here](#)

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 3 – Student registration form (UAT-A3).

CampusSafe Twin

LoginRegister



Student Registration

Join CampusSafe Twin to report incidents and access safe walking routes

Full Name

Thandi Mokoena

University / Student Email

thandi.uat@student.nwu.ac.za

Student Number

39884521

Password

Confirm Password

👁

Create Student Account

Already have an account? [Sign in here](#)

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 4 – Registration form completed for new student Thandi Mokoena ([UAT-A3](#)).

CampusSafe Twin

LoginRegister



CampusSafe Login

Access campus security and incident reporting

Registration successful! You may now sign in with your email and password.

Email Address

student@nwualum.co.za or admin@campussafe.ac.za

Password

Enter your password

Sign In

New student? [Create student account](#)

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 5 – Registration success confirmation on return to login ([UAT-A3](#)).

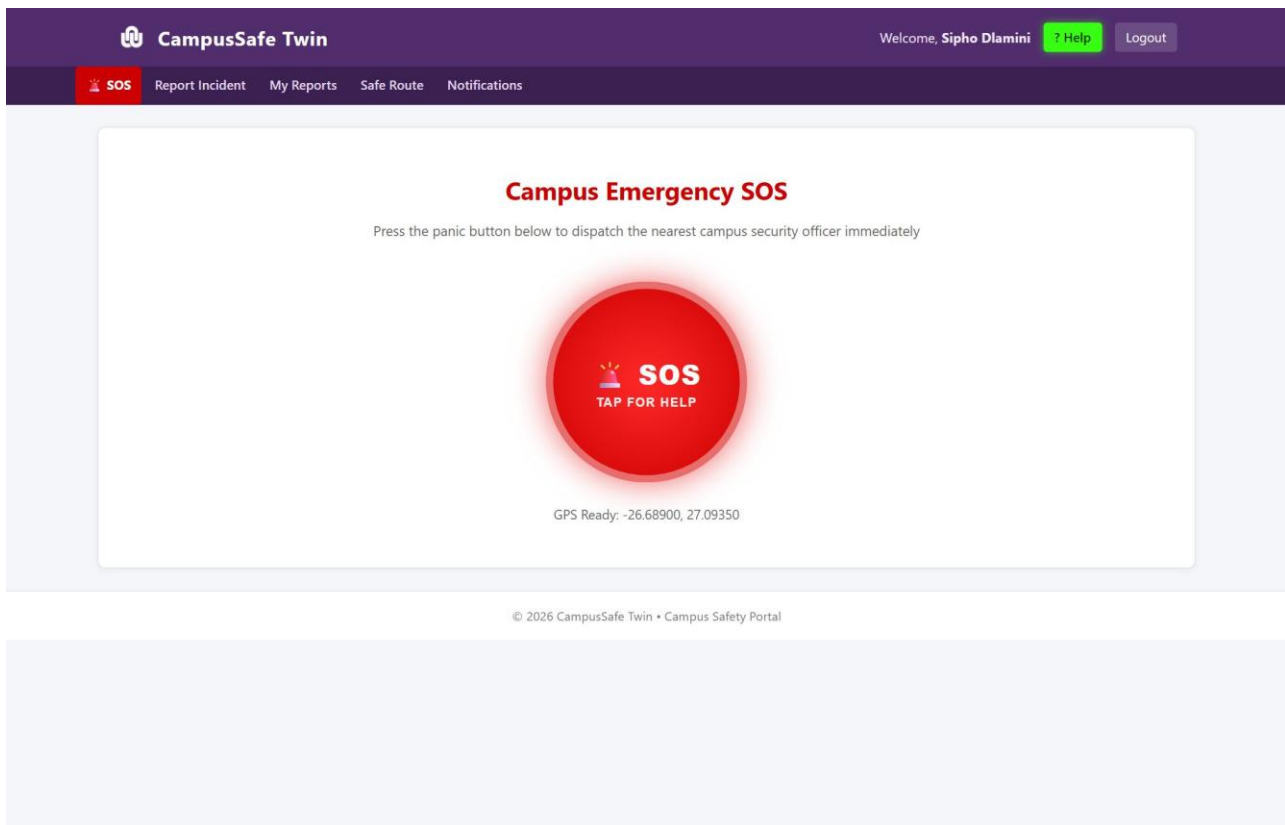


Figure 6 – Campus Emergency SOS page, student role ([UAT-B1](#), [UAT-A4](#)).

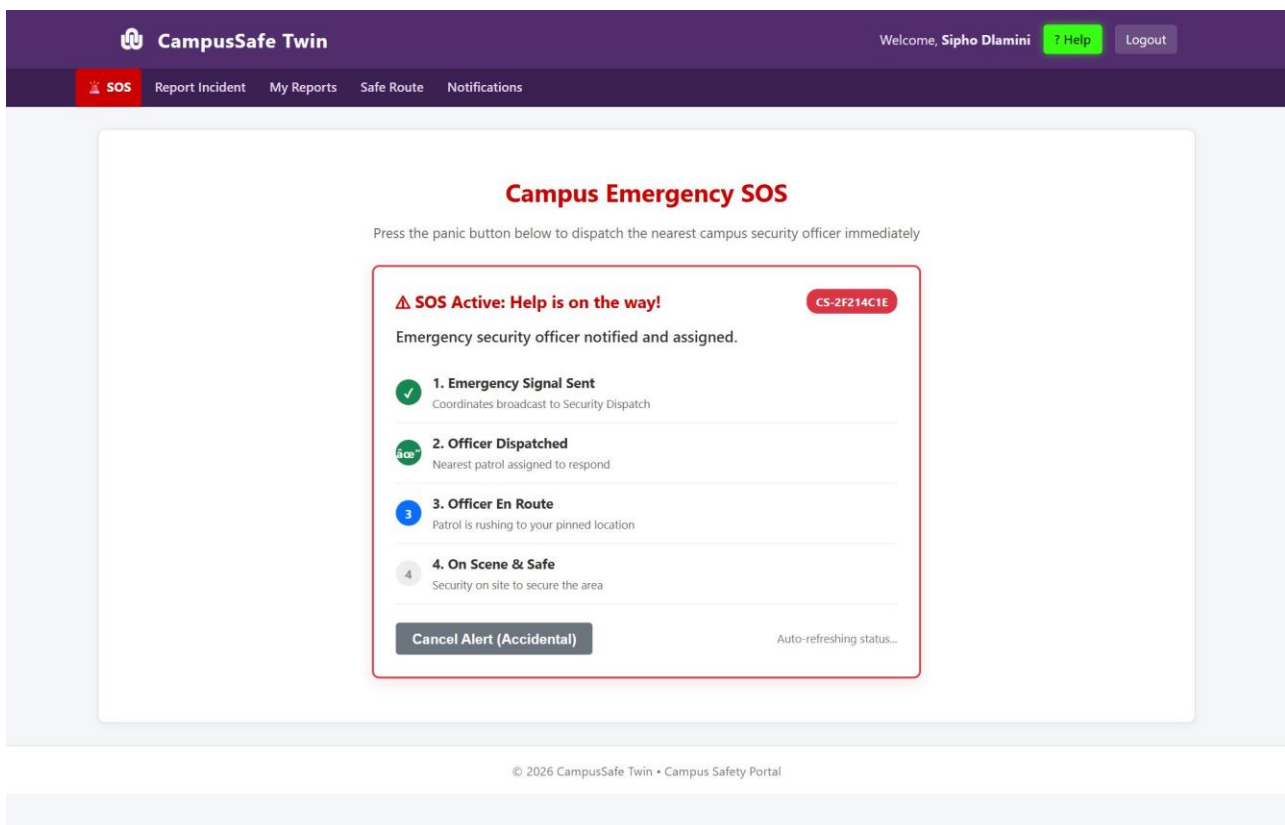


Figure 7 – SOS active: reference CS-2F214C1E, officer notified and assigned, live 4-stage tracker, cancel option ([UAT-B1](#); OBS-01 visible on stage-2 icon).

CampusSafe Twin

Welcome, Siphon Dlamini
? Help
Logout

SOS

Report Incident

My Reports

Safe Route

Notifications

Report a Safety Incident

Pin the location on the campus map and provide incident details to notify campus safety

Incident Category
Suspicious Activity

Date Happened
2026/10/05

Time Happened
19:30

Incident Description
UAT test report: Two unknown individuals loitering near the library bicycle racks after dark, trying bike locks. Wearing dark hoodies. Left towards the main gate when approached.

Pinned Location
Selected Coordinates: -26.689953, 27.093959

Evidence Photos (Up to 3, JPG/PNG, max 5 MB each):
Choose Files e9a030f3048b47f68fdc63fc766a883f.jpg

Optional photo evidence attached securely to your incident report.

Submit Incident Report

Map Location: Click anywhere on the campus map to drop or move your incident pin.

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 8 – Incident report completed: category, date/time, description, dropped map pin and photo evidence attached (UAT-B3).

CampusSafe Twin

Welcome, Siphon Dlamini
? Help
Logout

SOS

Report Incident

My Reports

Safe Route

Notifications

Report a Safety Incident

Pin the location on the campus map and provide incident details to notify campus safety

Report submitted successfully! Reference: CS-85BA6578

Incident Category
-- Select Category --

Date Happened
2026/10/05

Time Happened
19:30

Incident Description
Describe what occurred, any individuals involved, and relevant surroundings...

Pinned Location
Selected Coordinates: Not set, click map to pin

Evidence Photos (Up to 3, JPG/PNG, max 5 MB each):
Choose Files No file chosen

Optional photo evidence attached securely to your incident report.

Submit Incident Report

Map Location: Click anywhere on the campus map to drop or move your incident pin.

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 9 – Report submission confirmation for CS-85BA6578 (UAT-B3).

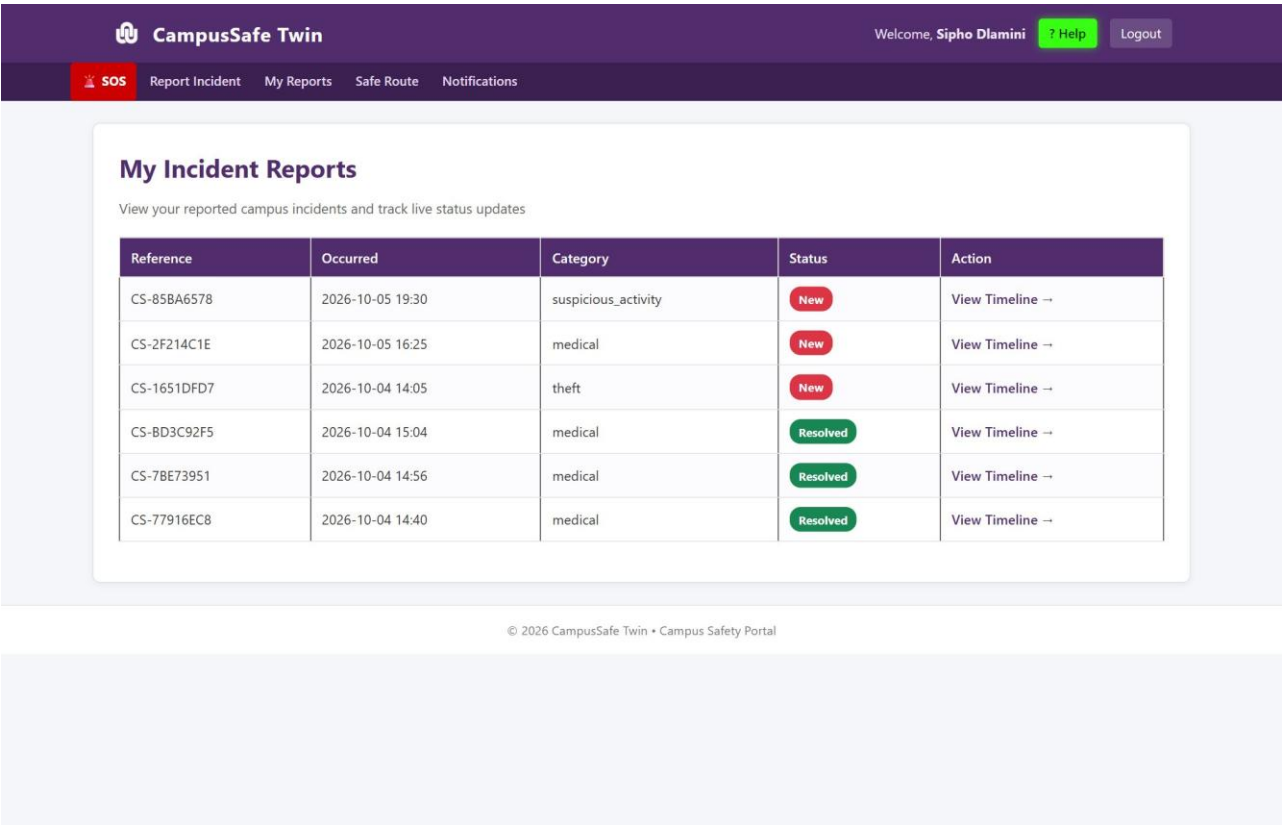


Figure 10 – My Incident Reports list with status badges and timeline action (UAT-B4).

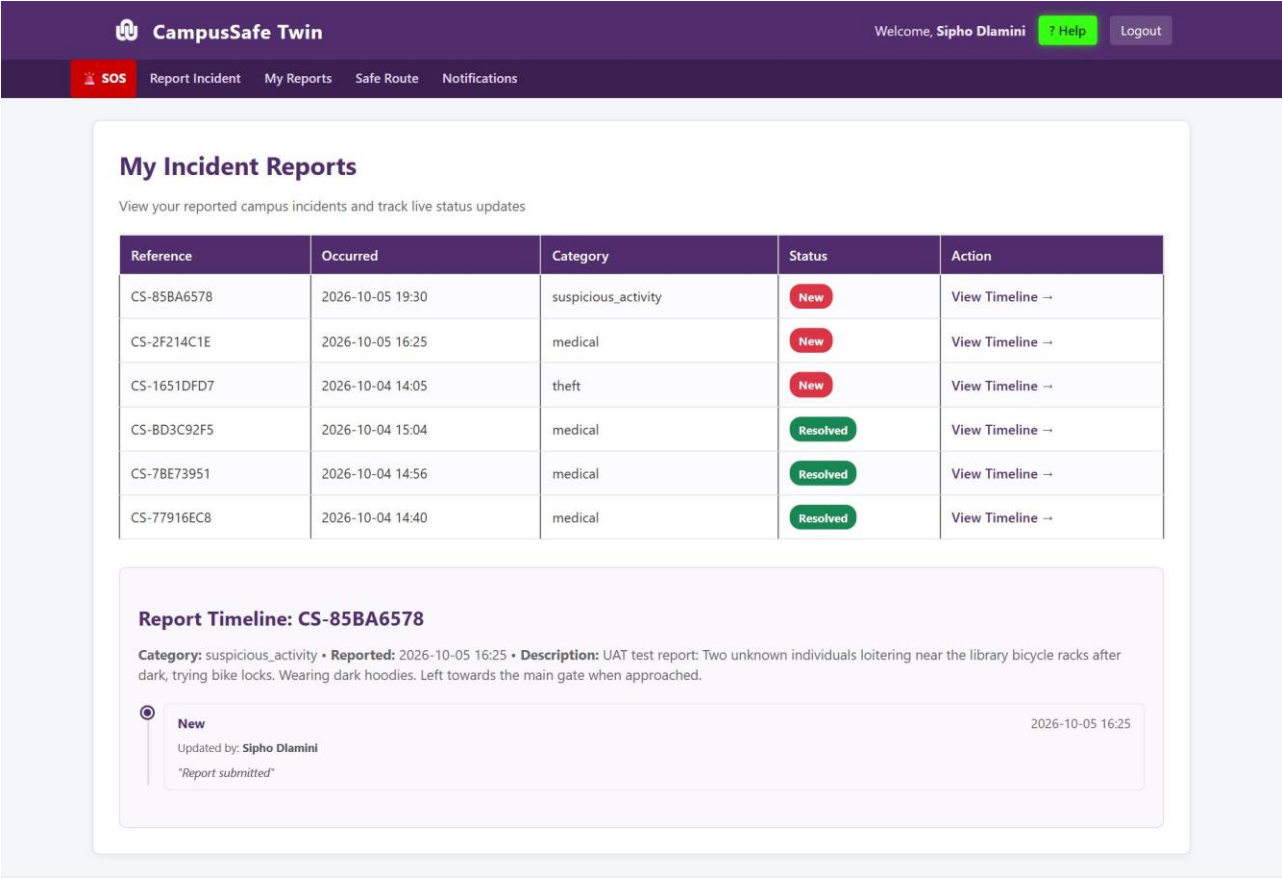


Figure 11 – Report status timeline for a selected report (UAT-B5).

Safer Campus Walking Route

Dynamic Dijkstra path routing incorporating historical incident proximity, lighting conditions, and active flagged zones

Select Campus Endpoints

From (Start Location):

B6

To (Destination Location):

E6 JS van der Merwe-gebou

Find Safest Route

↺ Clear Map

Or click anywhere on the campus map below to drop custom Start (A) and Destination (B) pins.

Route calculated successfully!

 Recommended Safer Route

Minimizes proximity to recent incidents and avoids dark unlit pathways

Distance

623.8 m

Risk score: 0

Est. Walking Time (1.3 m/s)

8 min

Shortest Direct Route

Direct geometric shortest path without risk weighting

Distance

623.8 m

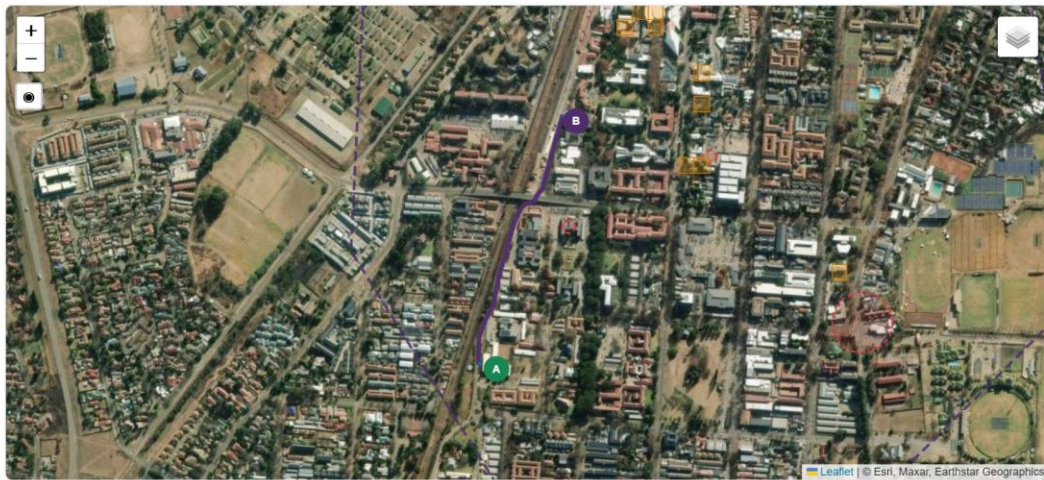
Est. Walking Time (1.3 m/s)

8 min

Save this route for quick access & condition alerts:

e.g. Evening walk from Library to Residence

Save Route



Solid Purple: Safer Recommended Walking Route

— **Dashed Grey:** Shortest Route Along Walkways

— **Red Circles:** Flagged Unsafe Zones

— **Orange Cells:** Incident Hotspots

My Saved Routes

No saved routes yet. Pick points above and click 'Save Route'.

Figure 12 – Safe route computed between campus locations: 623.8 m / 8 min, route drawn A→B on satellite map with risk overlays and save option ([UAT-B6](#)).

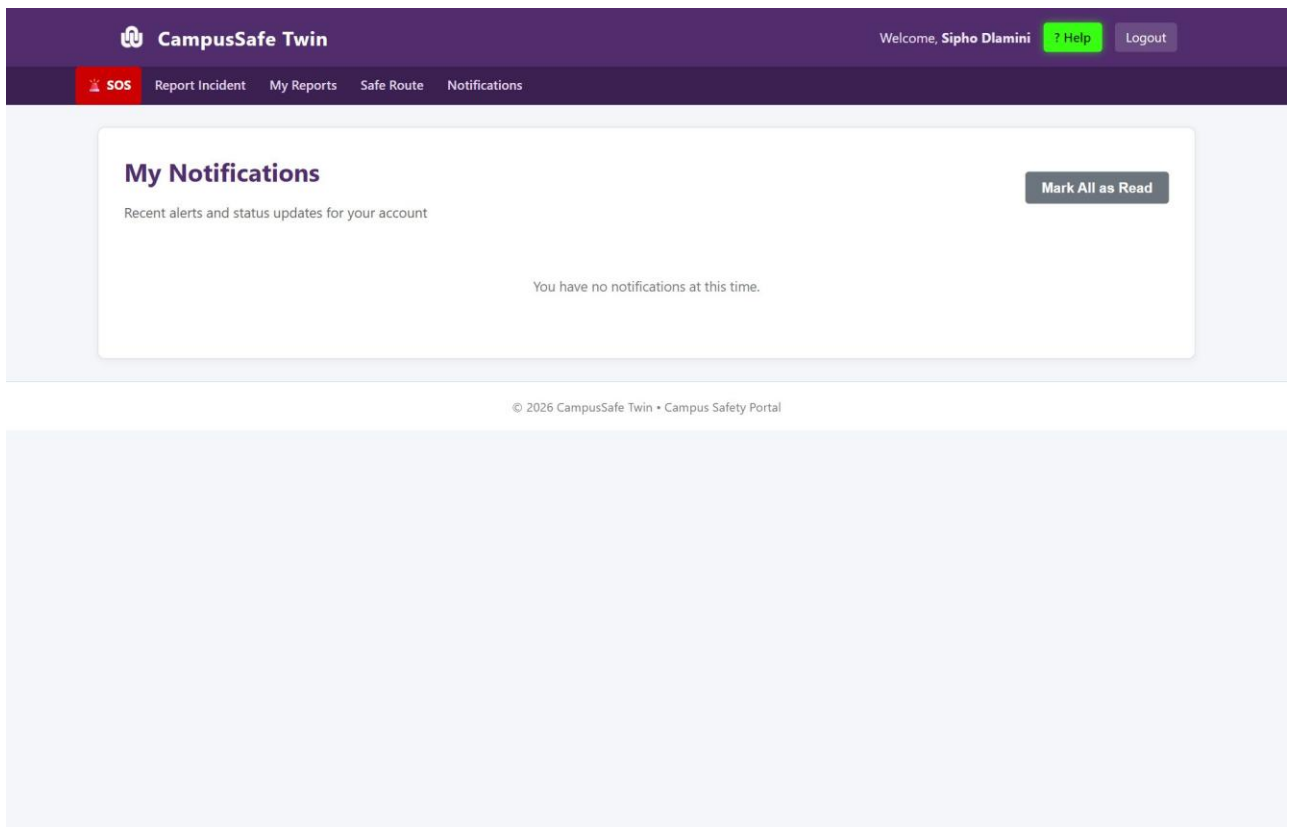


Figure 13 – My Notifications with Mark All as Read ([UAT-B7](#)).

CampusSafe Twin

Welcome, Siphon Dlamini
? Help
Logout

SOS

Report Incident

My Reports

Safe Route

Notifications

Help Centre

Not sure how something works? Search the user guides below and open the PDF that covers the part of the system you need help with.

Search

Show All

Showing all 7 help guide(s).

ADMIN

Supervisor & Administration Guide

For supervisors: live ops, hotspots, managing staff and locations, analytics, and uploading help guides.

PDF • 5 KB • Added 04 Oct 2026

Open Guide

GETTING STARTED

Getting Started with CampusSafe Twin

How to register an account, log in, and find your way around the portal.

PDF • 5 KB • Added 04 Oct 2026

Open Guide

NOTIFICATIONS

Tracking My Reports & Notifications

How to follow up on reports you submitted and understand the notifications the system sends you.

PDF • 4 KB • Added 04 Oct 2026

Open Guide

REPORTING

How to Report an Incident

Step by step guide to reporting theft, harassment, suspicious activity and other incidents with photos and a map pin.

PDF • 5 KB • Added 04 Oct 2026

Open Guide

SAFE ROUTE

Finding a Safe Route Across Campus

How to plan a walking route that avoids recent incident hotspots using the Safe Route planner.

PDF • 4 KB • Added 04 Oct 2026

Open Guide

SECURITY OPS

Security Officer Field Guide

For security staff: using the live security map, responding to dispatches, and flagging risky zones.

PDF • 5 KB • Added 04 Oct 2026

Open Guide

SOS

Using the SOS Emergency Button

What the red SOS button does, when to use it, and what happens after you press it.

PDF • 5 KB • Added 04 Oct 2026

Open Guide

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 14 – Help Centre with search (UAT-B8).

 CampusSafe Twin

Welcome, Sipho Dlamini [? Help](#) [Logout](#)

 SOS

[Report Incident](#)

[My Reports](#)

[Safe Route](#)

[Notifications](#)

Report a Safety Incident

Pin the location on the campus map and provide incident details to notify campus safety

Incident Category

-- Select Category --

Date Happened

2026/10/05

Time Happened

16:26

Incident Description

Describe what occurred, any individuals involved, and relevant surroundings...

Pinned Location

Selected Coordinates: **Not set, click map to pin**

Evidence Photos (Up to 3, JPG/PNG, max 5 MB each):

Choose Files

No file chosen

Optional photo evidence attached securely to your incident report.

Submit Incident Report

Map Location: Click anywhere on the campus map to drop or move your incident pin.



Figure 15 – Access control: student's direct request for /Admin/Users.aspx redirected to the student dashboard ([UAT-A5](#)).

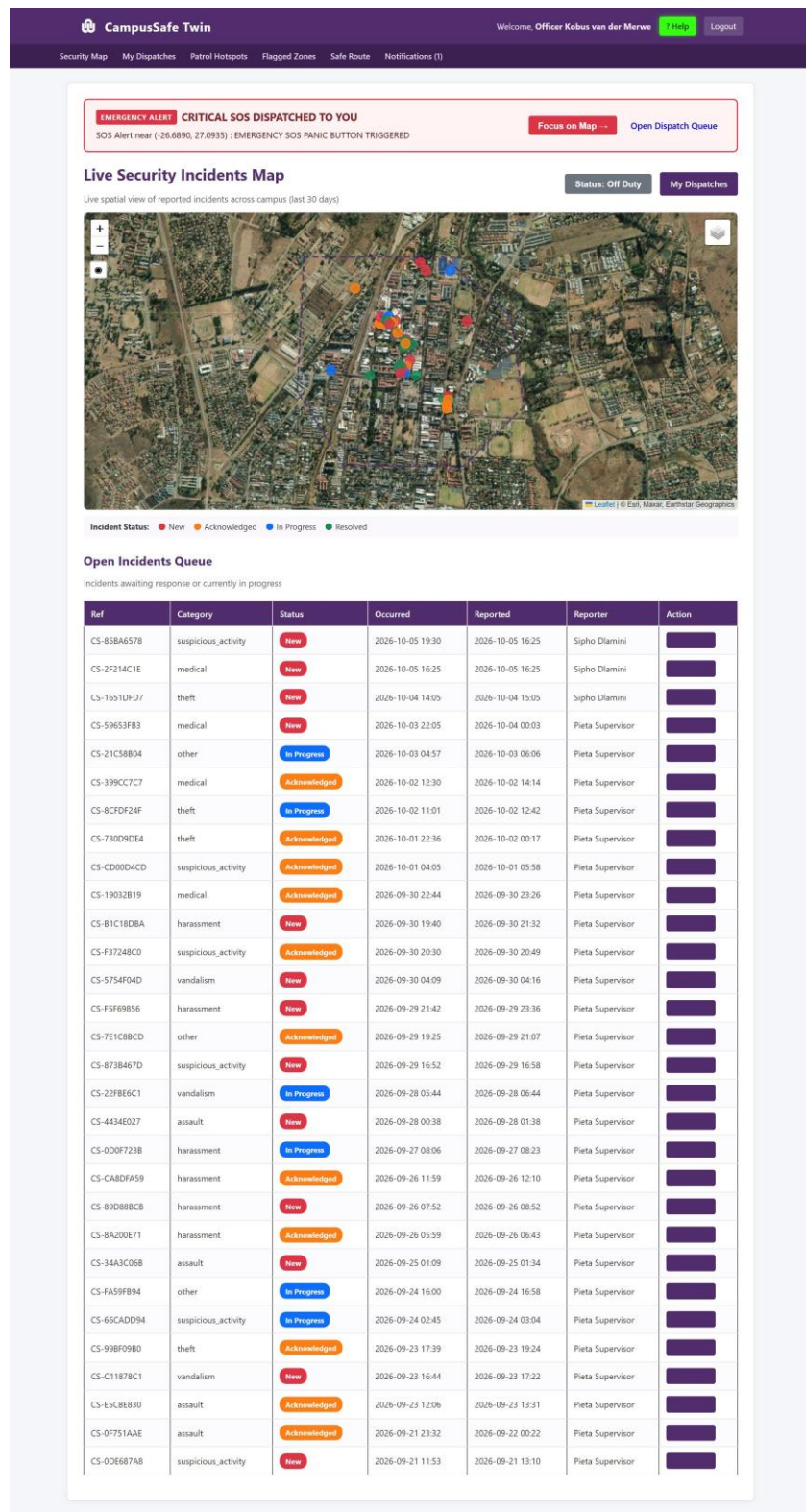


Figure 16 – Guard: Live Security Incidents Map with critical SOS banner, colour-coded markers, status legend and Open Incidents Queue including UAT-created records CS-85BA6578 and CS-2F214C1E ([UAT-C1](#), [UAT-B2](#); OBS-02 visible in Action column).

CampusSafe Twin

Welcome, **Officer Kobus van der Merwe**
[? Help](#)
[Logout](#)

[Security Map](#)
[My Dispatches](#)
[Patrol Hotspots](#)
[Flagged Zones](#)
[Safe Route](#)
[Notifications \(1\)](#)

Incident Details: CS-21C58B04

Reference: 21c58b04-c7cf-4c7d-98aa-af50f48b2dfb

[← Back to Security Map](#)

Incident Information

Status	In Progress
Category	other
Occurred	2026-10-03 04:57
Reported	2026-10-03 06:06
Reporter	Pieta Supervisor
Coordinates	-26.690429, 27.093511

Description:

Student bag reported stolen from bench near library steps.

Incident Location Pin

Status History Timeline

Evidence Photos & Notes

No photo evidence attached.

Staff Notes:

Add Investigation Note:

Add note or observation... [Add Note](#)

Lifecycle Action

Move incident to next status: **Resolved**

Action Note (Optional):

e.g. Dispatched officer or resolved on scene...

[Mark Resolved](#)

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 17 – Incident detail CS-21C58B04: information panel, location pin, notes and lifecycle action to "Mark Resolved" ([UAT-C2](#)).

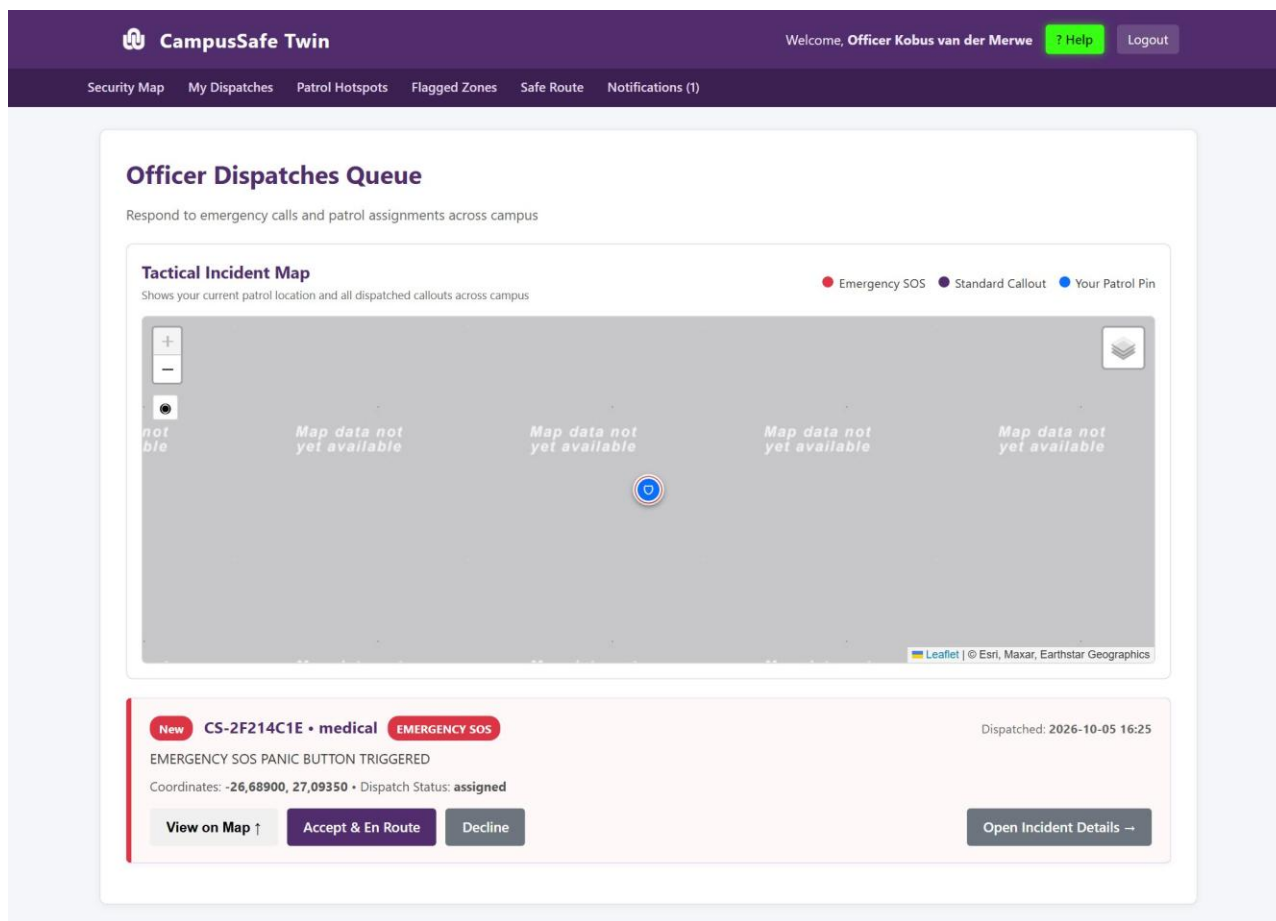
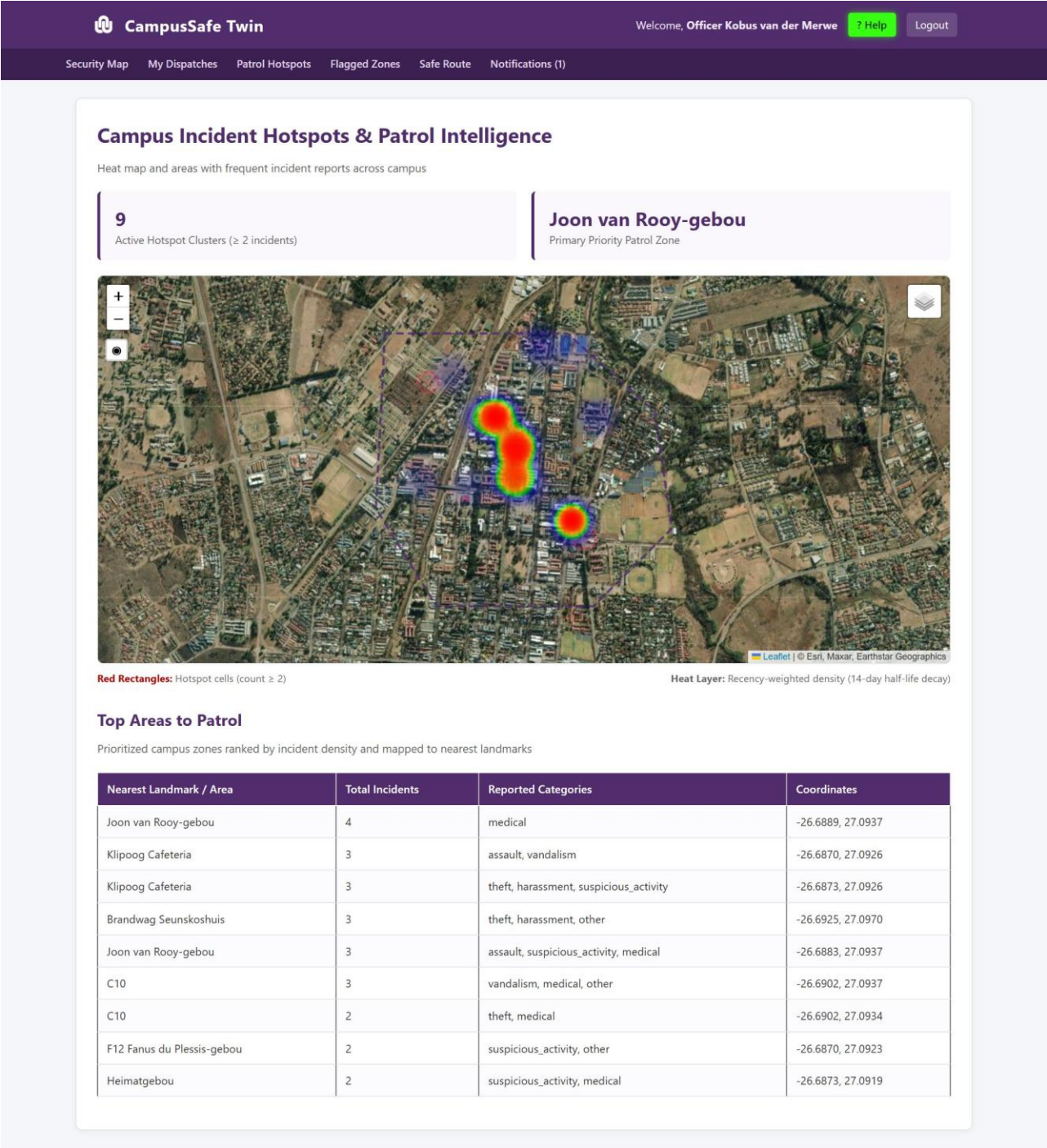


Figure 18 – Officer Dispatches Queue: EMERGENCY SOS dispatch CS-2F214C1E assigned with Accept & En Route / Decline actions and tactical map ([UAT-C3](#), [UAT-B2](#); OBS-03 visible on map tiles).



Flagged Safety Zones

Designate hazard areas and general campus risk perimeters that safe walking routes must actively avoid

Add New Flagged Zone

Click the map to place the zone center.

Zone Name

e.g. Construction Area / Poor Lighting

Risk Severity (1 = Low, 5 = Critical)

Level 3 - Moderate Risk

Radius (Metres)

50

Reason / Notes

e.g. Unlit path, reported muggings...

Center Coordinates

Click map to set location

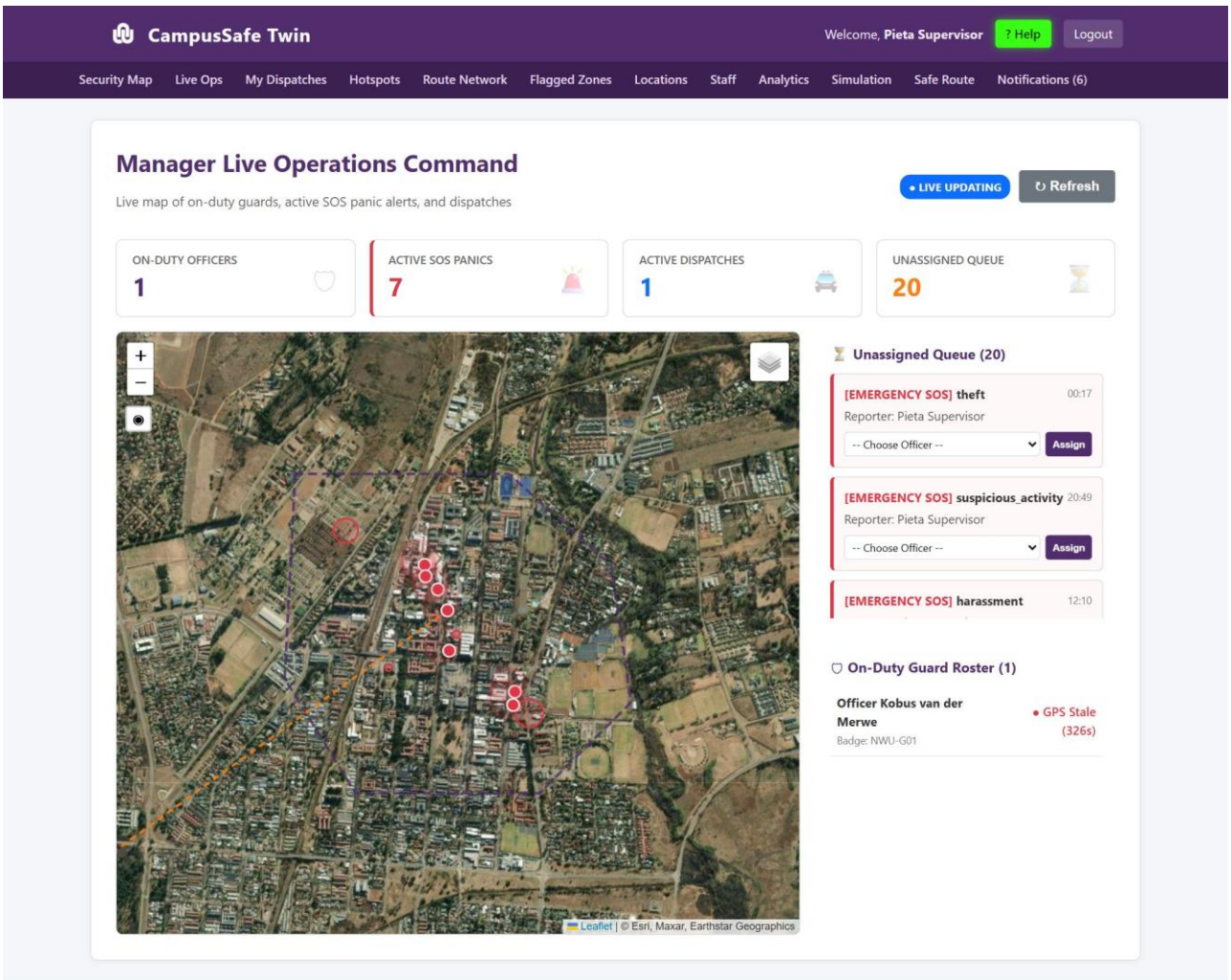
Create Flagged Zone



Active Zones

Zone	Risk	Radius	Action
Serious issue	5	15	Remove
Car Crasg	2	15	Remove
Construction site - reduced visibility	2	60	Remove
Dimly lit parking area	3	70	Remove

Figure 20 – Flagged Safety Zones: zone list, map and creation form ([UAT-C5](#)).



© 2026 CampusSafe Twin • Campus Safety Portal

Figure 21 – Supervisor: Manager Live Operations Command (UAT-D1).

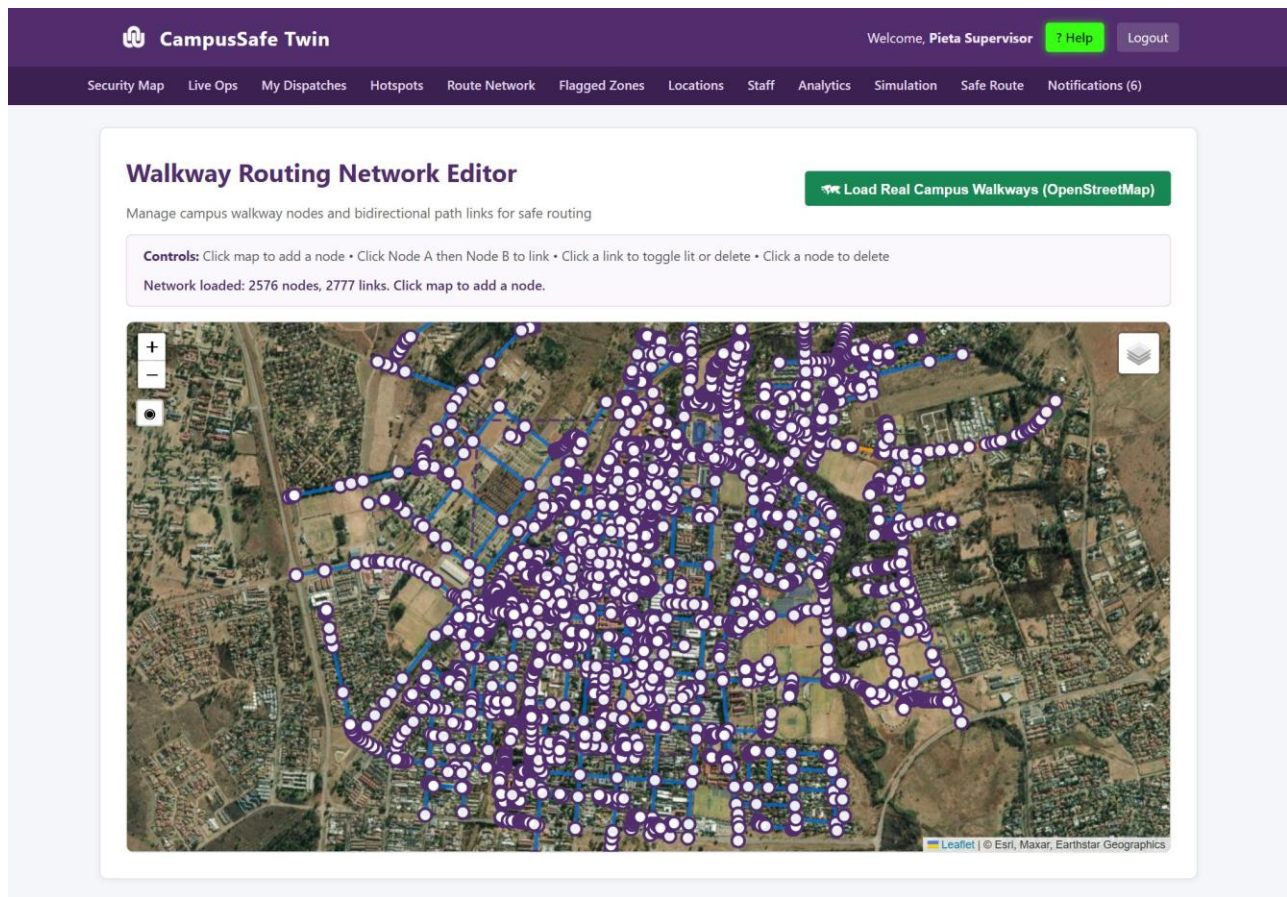


Figure 22 – Walkway Routing Network Editor rendering the OSM-derived network ([UAT-D2](#)).

Welcome, Pieta Supervisor

[Help](#)
[Logout](#)

[Security Map](#)
[Live Ops](#)
[My Dispatches](#)
[Hotspots](#)
[Route Network](#)
[Flagged Zones](#)
[Locations](#)
[Staff](#)
[Analytics](#)
[Simulation](#)
[Safe Route](#)
[Notifications \(6\)](#)

Campus Locations & Landmarks

Manage Points of Interest (POIs) used for Safe Route departure/destination points and hotspot references

[Seed Standard Campus POIs](#)

Active Campus Landmarks (60)

Name	Category	Coordinates	Action
B6	building	26.69040, 27.09090	Delete
B9	building	26.69040, 27.09040	Delete
Bonjour	building	26.69074, 27.09000	Delete
Brandwag Seunskoshuis	building	26.69064, 27.09070	Delete
C10	building	26.69037, 27.09044	Delete
C1: Institutional Office	building	26.69090, 27.09060	Delete
C3B	building	26.69017, 27.09040	Delete
C3C	building	26.69036, 27.09090	Delete
C6	building	26.69004, 27.09070	Delete
C9	building	26.69045, 27.09060	Delete
Cafe Fiori	building	26.69060, 27.09060	Delete
D1	building	26.69006, 27.09040	Delete
D2	building	26.69009, 27.09040	Delete
Die Akker	building	26.69442, 27.09077	Delete
Drakenstein	building	26.69020, 27.09074	Delete
E6 JS van der Merwe-gebou	building	26.69020, 27.09074	Delete
Equilibria School of Life	building	26.69091, 27.09070	Delete
F12 Fanus du Plessis-gebou	building	26.69080, 27.09090	Delete
F5 Ou Hooftgebou	building	26.69067, 27.09090	Delete
F6 Totiusaal	building	26.69019, 27.09074	Delete
Fanie du Toit Rugby Veld	sports	26.69090, 27.09090	Delete
Ferdinand Postma-biblioteek	building	26.69019, 27.09070	Delete
G1	building	26.69088, 27.09010	Delete
G17	building	26.69024, 27.09090	Delete
G20	building	26.69010, 27.09074	Delete
G4	building	26.69032, 27.09090	Delete
G5	building	26.69039, 27.09070	Delete
G6	building	26.69030, 27.09010	Delete
G7	building	26.69090, 27.09090	Delete
G8	building	26.69004, 27.09080	Delete
Heimatgebou	building	26.69032, 27.09090	Delete
Hello Sushi	building	26.69044, 27.09080	Delete
J6	building	26.69090, 27.09080	Delete
J8 Marks Oval	sports	26.69044, 27.09090	Delete
Joon van Rooy-gebou	building	26.69020, 27.09074	Delete
Klippoeg Cafeteria	building	26.69070, 27.09074	Delete
Klubhuis	building	26.69040, 27.09070	Delete
Koshuis Vader Huis	building	26.69047, 27.09060	Delete
Laarskool Mooinvier	building	26.69074, 27.09070	Delete
N2A	building	26.69040, 27.09070	Delete
N2B	building	26.69074, 27.09090	Delete
N3A	building	26.69042, 27.09040	Delete
N3B	building	26.69041, 27.09040	Delete
N3C	building	26.69042, 27.09040	Delete
N3D	building	26.69090, 27.09090	Delete
NWU	building	26.69042, 27.09090	Delete
NWU Library Grass	other	26.69072, 27.09070	Delete
NWU-Puk Astro	sports	26.69045, 27.09070	Delete
North West University	building	26.69030, 27.09090	Delete
North-West University, Sports Centre	building	26.69020, 27.09070	Delete
Sanlam Auditorium	building	26.69060, 27.09090	Delete
Shell Select	building	26.69067, 27.09070	Delete
Soetdoring's Mans Koshuis	building	26.69044, 27.09090	Delete
Soetdoring's Menskoshuis	building	26.69044, 27.09090	Delete
Steers	building	26.69011, 27.09044	Delete
Strydom House	building	26.69090, 27.09090	Delete
Superspar	building	26.69092, 27.09040	Delete
Toro	building	26.69070, 27.09060	Delete
Villagers Klubhuis	building	26.69060, 27.09090	Delete
deMILKO	building	26.69070, 27.09030	Delete

Add New Campus POI

Location / Building Name:

e.g. Ferdinand Postma Library

Category:

Academic / Building

Description (Optional):

e.g. North campus gate opposite student centre

Latitude: Longitude:

-26.6900 27.0905

Click anywhere on the mini map to auto-fill latitude and longitude:

Save Campus Location

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 23 – Campus Locations & Landmarks management (UAT-D3).

 CampusSafe Twin

Welcome, Pieta Supervisor Help Logout

[Security Map](#) [Live Ops](#) [My Dispatches](#) [Hotspots](#) [Route Network](#) [Flagged Zones](#) [Locations](#) [Staff](#) [Analytics](#) [Simulation](#) [Safe Route](#) [Notifications \(6\)](#)

User Accounts & Staff Access

Manage campus security guards, supervisors, and student accounts, role permissions, and access status

Registered Accounts (4)

All Roles

Full Name	Email	Role	Identifier	Active Status	Actions
Officer Kobus van der Merwe	kobus@guard.nwu.ac.za		Badge: NWU-G01	Active	Deactivate Reset PW
Sipho Dlamini	sipho@student.nwu.ac.za		Std: 29384721	Active	Deactivate Reset PW
Campus Supervisor	supervisor@campussafe.ac.za		-	Active	Deactivate Reset PW
Pieta Supervisor	pieta@security.nwu.ac.za		Badge: NWU-S01	Active	Deactivate Reset PW

Create New Account

Full Name:

e.g. Officer John Smith

Email Address:

officer@nwu.ac.za

Role:

Patrol Guard

Badge Number / Student ID:

e.g. NWU-G44

Phone Number:

018 299 1111

Temporary Password:

CampusSafe2026!

Create Account

© 2026 CampusSafe Twin • Campus Safety Portal

Figure 24 – User Accounts & Staff Access administration (UAT-D4).

Export All Incidents (CSV)

Overview of safety metrics, incident types and officer response times

N/A

Category	Incidents	Percentage
MEDICAL	10 incidents	(22%)
HARASSMENT	8 incidents	(17%)
SUSPICIOUS ACTIVITY	7 incidents	(15%)
THEFT	6 incidents	(13%)
ASSAULT	6 incidents	(13%)
VANDALISM	5 incidents	(11%)
OTHER	4 incidents	(9%)

New	13 incidents (28%)
Acknowledged	12 incidents (26%)
In Progress	6 incidents (13%)
Resolved	15 incidents (33%)

Ref / Dispatch	Type	Responding Guard	Dispatch Status	Dispatched At	Time to Arrive	Time to Resolve
f7860654	[SOS EMERGENCY] medical	Officer Kobus van der Merwe	assigned	2026-10-05 16:25	Pending	In Progress
272812ae	[SOS EMERGENCY] medical	Auto-assigned	cancelled	2026-10-04 15:04	Pending	0,2 min
ee987a58	[SOS EMERGENCY] medical	Officer Kobus van der Merwe	cancelled	2026-10-04 14:56	Pending	0,1 min
ca9c804d	[SOS EMERGENCY] medical	Officer Kobus van der Merwe	cancelled	2026-10-04 14:40	Pending	0,0 min

Figure 25 – Campus Safety Intelligence & Analytics: KPIs (46 incidents, 14 SOS), category and resolution distributions, dispatch history, CSV export ([UAT-D5](#); OBS-04 visible in KPI tiles).

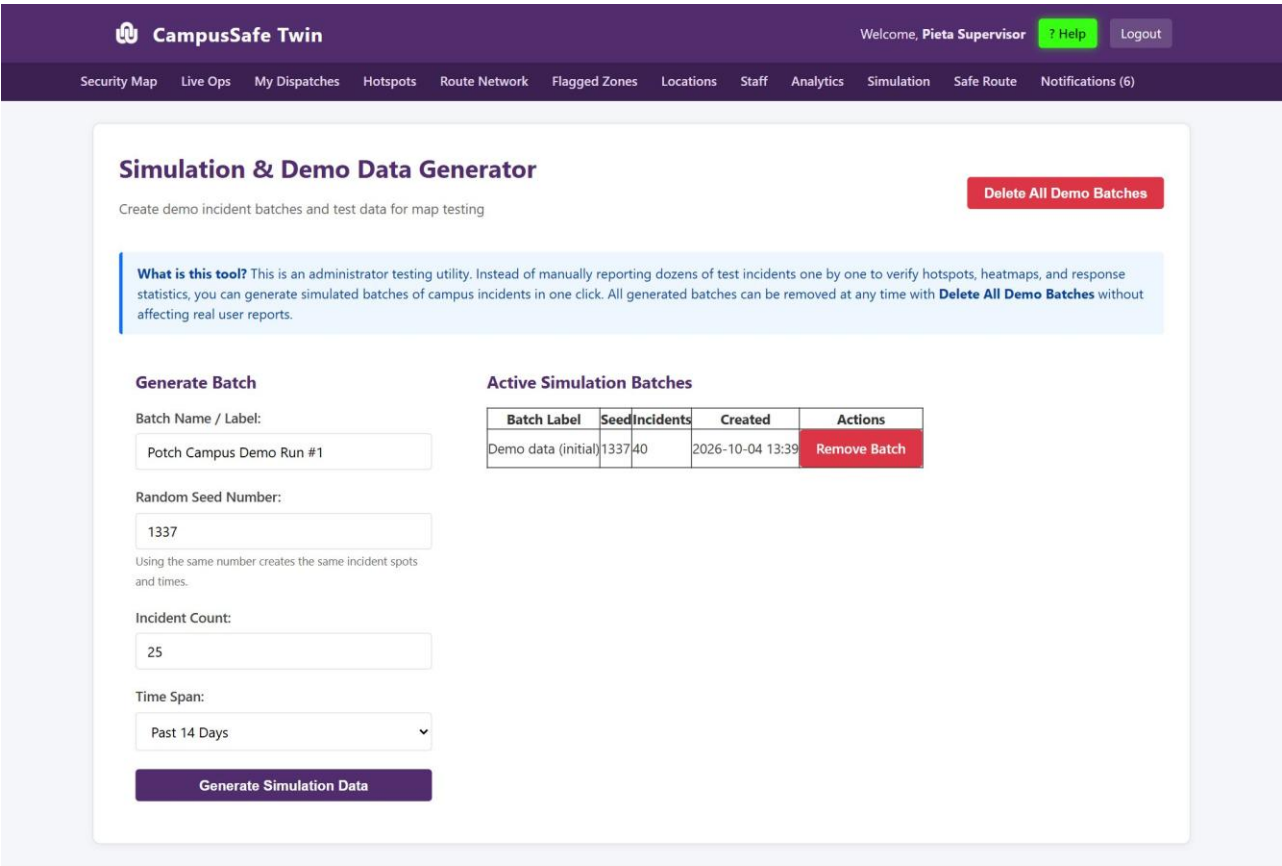


Figure 26 – Simulation & Demo Data Generator with guarded purge controls (UAT-D6).

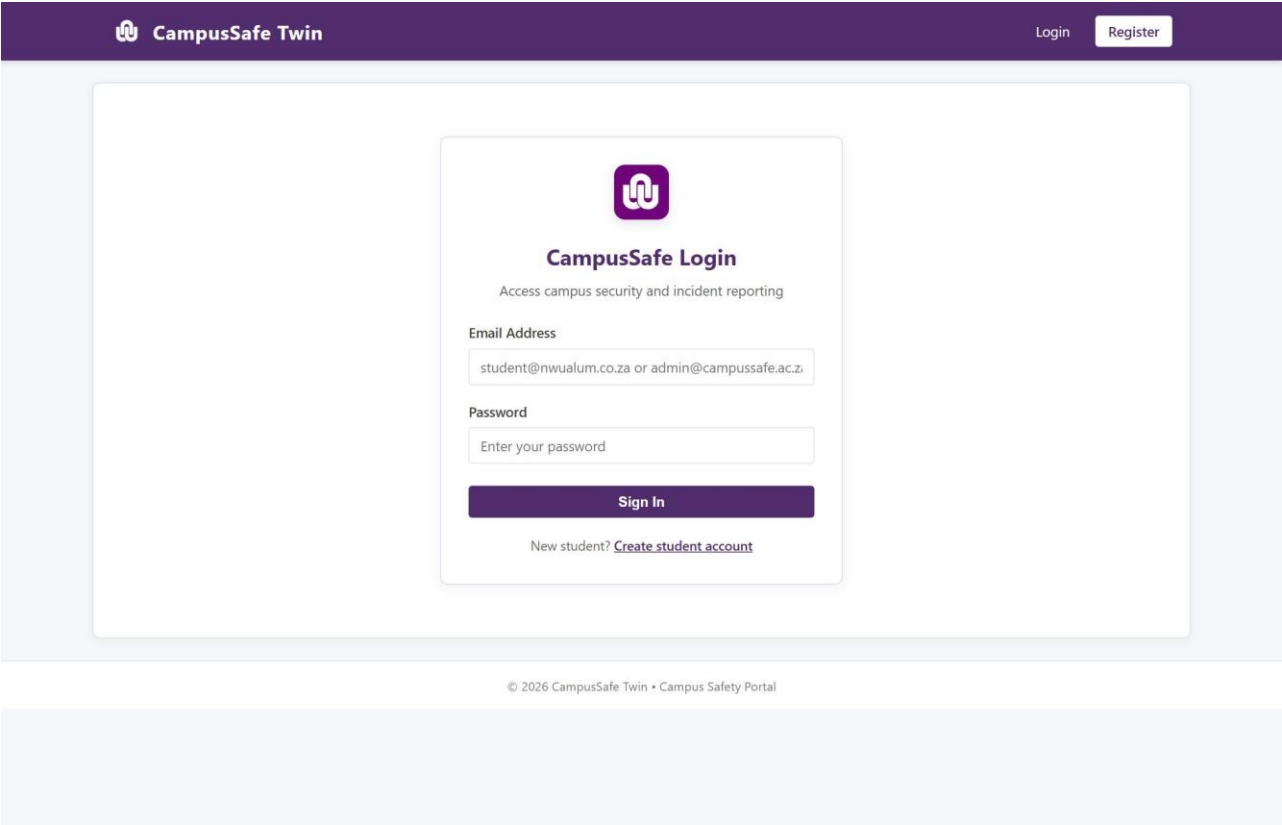


Figure 27 – Session ended: returned to login in signed-out state (UAT-A6).

7. Observation & Defect Log

No Severity 1 or 2 defects were found. Four low-severity observations were logged during Round 1 execution; none blocked a business workflow. The table below records each finding as observed, with the remediation recommendation made at the time. **All four observations have since been fixed and re verified**, see [Section 7.1](#).

ID	Severity	Description & recommendation	Evidence
OBS-01	Sev 4	SOS tracker icon garbled. Stage 2 ("Officer Dispatched") of the SOS progress tracker renders the characters "âœ“" instead of a check mark: a UTF-8 check-mark glyph decoded with the wrong encoding. Recommendation: emit the icon as an HTML entity (✓) or ensure the response/page encoding is UTF-8.	Fig. 7
OBS-02	Sev 4	Unlabelled queue action buttons. The per-row "Action" buttons in the Security Map Open Incidents Queue render as plain purple rectangles with no visible caption. The action itself works (incident detail reachable). Recommendation: verify the button glyph/label renders in all browsers; add a text label or tooltip.	Fig. 16
OBS-03	Sev 3	Tactical map imagery unavailable at default zoom. On My Dispatches the tactical incident map displayed "Map data not yet available" tiles at its initial zoom level during the session (markers and patrol pin still rendered; zooming out restores imagery). Likely the Esri satellite layer has no imagery at that deep zoom for this area. Recommendation: cap the initial/max zoom to the deepest level with available imagery, or fall back to the street base layer.	Fig. 18
OBS-04	Sev 4	Response-time KPIs show "N/A". Analytics shows "N/A" for average response and resolution time although dispatch history rows exist; arrivals in the current dataset are pending, so no averages can yet be computed. Behaviour is arguably correct but may confuse managers. Recommendation: display "No completed dispatches yet" instead of "N/A", and confirm the calculation populates once officers log arrival.	Fig. 25

7.1 Remediation & Re-verification (03 October 2026, post-Round 1)

The development team implemented fixes for all four observations on the day of execution. Each fix was re-verified in the live UAT environment by Maseko S with the same scripted-browser method used for Round 1; the after-fix evidence is reproduced in Figures 28–31. The Round 1 evidence figures (7, 16, 18, 25) are retained unchanged as the "before" record.

ID	Fix implemented	Re-verification result	Evidence
OBS-01	The SOS status script no longer emits a raw check-mark character (vulnerable to encoding mismatch); it now sets the icon via a character-code call (Student/Sos.aspx), which is encoding-safe.	A fresh SOS was triggered: the completed step icons render as a correct check mark (verified programmatically as code point 10003). The test alert was cancelled afterwards via Cancel Alert (Accidental) .	Fig. 28
OBS-02	A stylesheet rule (.grid-view a) was overriding the button text colour inside data grids, producing purple-on-purple. A more specific rule (.grid-view a.btn, Site.css) now keeps button text white.	The "Manage →" labels in the Security Map Open Incidents Queue are clearly legible; the computed text colour on the live page is white (rgb 255, 255, 255).	Fig. 29
OBS-03	The satellite base layer is now capped at its deepest native imagery level (maxNativeZoom: 18, Scripts/campusmap.js), with deeper zooms upscaled instead of requesting unavailable tiles; the tactical map's auto-fit is additionally capped at zoom 17 (Security/MyDispatches.aspx).	The My Dispatches tactical map renders full satellite imagery at its default view where it previously showed "Map data not yet available" placeholder tiles.	Fig. 30
OBS-04	When no dispatch has a logged arrival yet, the Analytics response-time KPIs now display "No completed dispatches yet" instead of "N/A" (Admin/Analytics.aspx.cs).	Both KPI tiles display the explanatory message, correctly sized within the tile.	Fig. 31

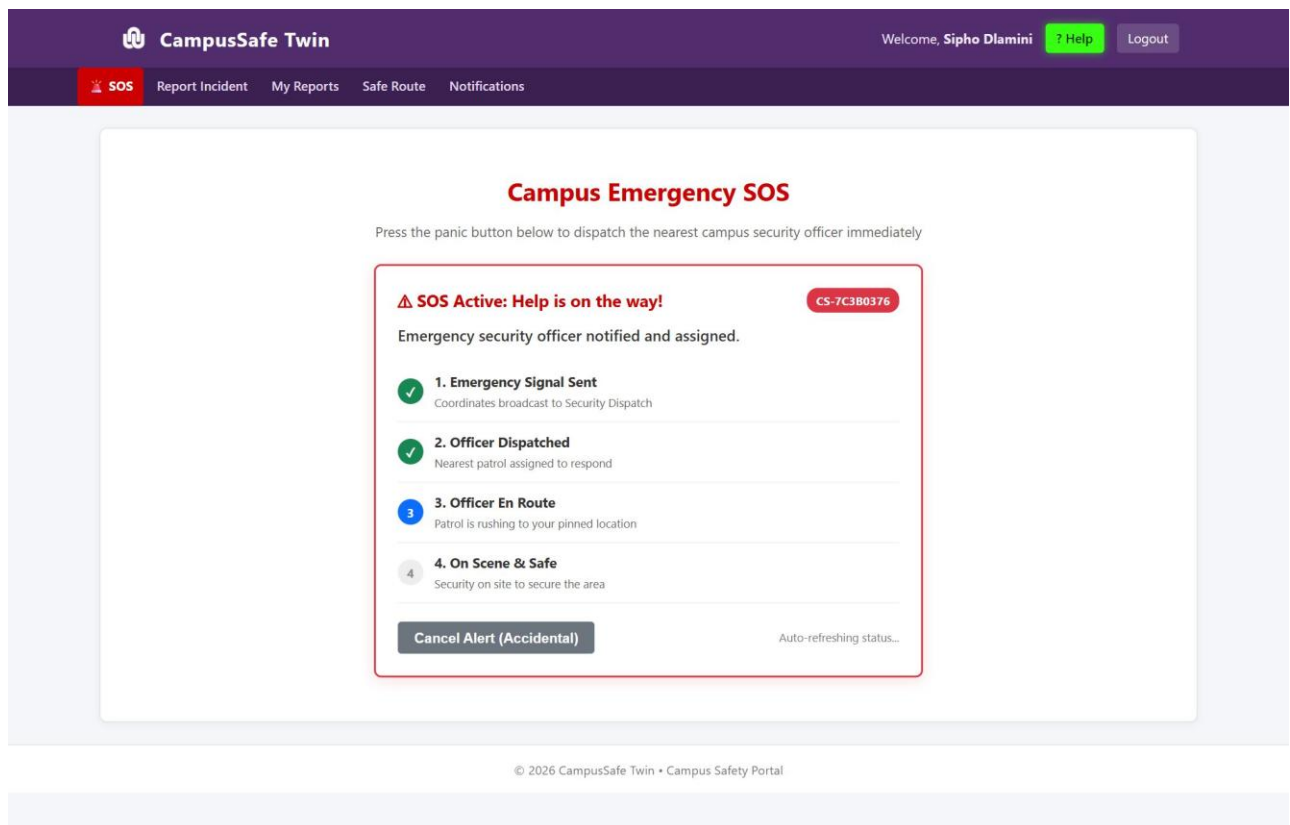


Figure 28 – After fix OBS-01: SOS re-triggered for verification; completed tracker steps show correct check marks (compare Figure 7).

CS-399CC/L7	medical	Acknowledged	2026-10-02 12:30	2026-10-02 14:14	Pieta Supervisor	Manage →
CS-8CFDF24F	theft	In Progress	2026-10-02 11:01	2026-10-02 12:42	Pieta Supervisor	Manage →
CS-730D9DE4	theft	Acknowledged	2026-10-01 22:36	2026-10-02 00:17	Pieta Supervisor	Manage →
.....						

Figure 29 – After fix OBS-02: "Manage →" action buttons in the Open Incidents Queue now legible in white (compare [Figure 16](#)).

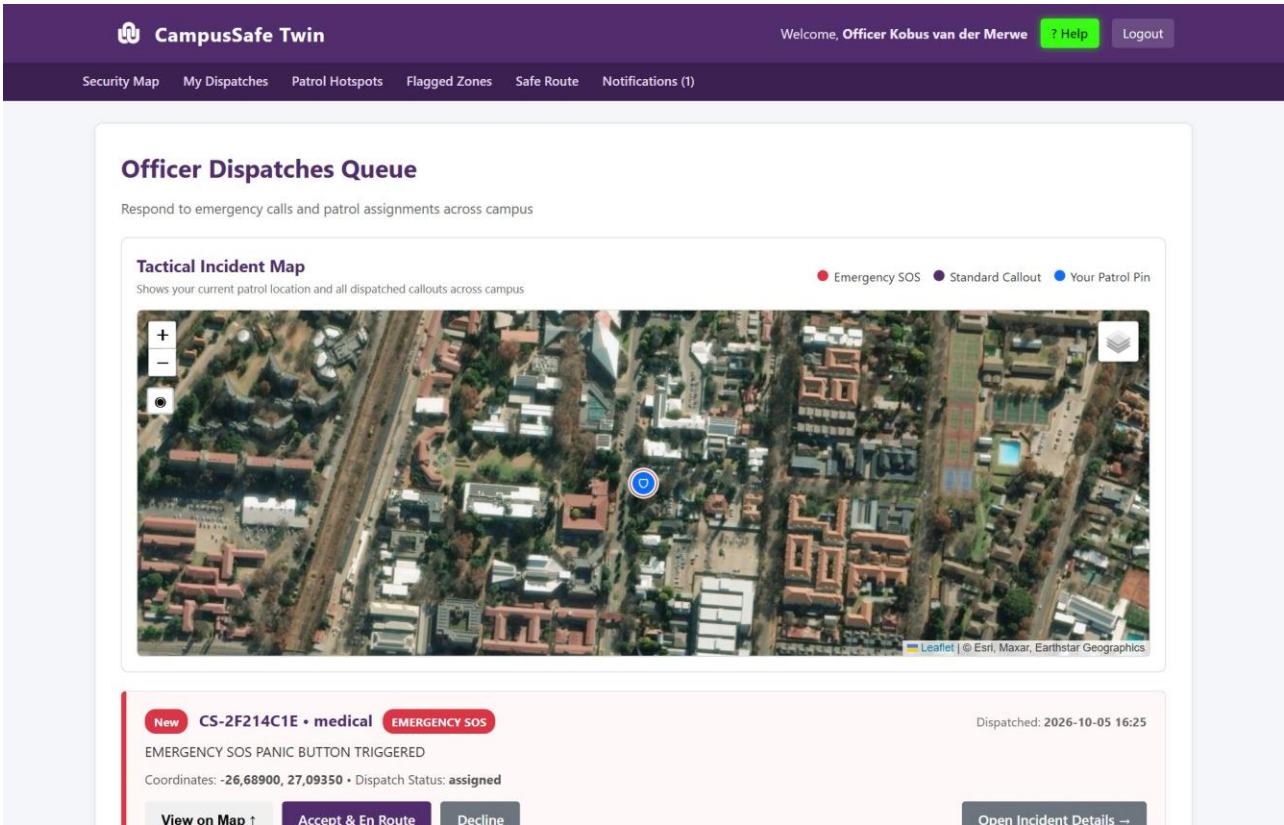


Figure 30 – After fix OBS-03: My Dispatches tactical map rendering full satellite imagery at its default view (compare [Figure 18](#)).

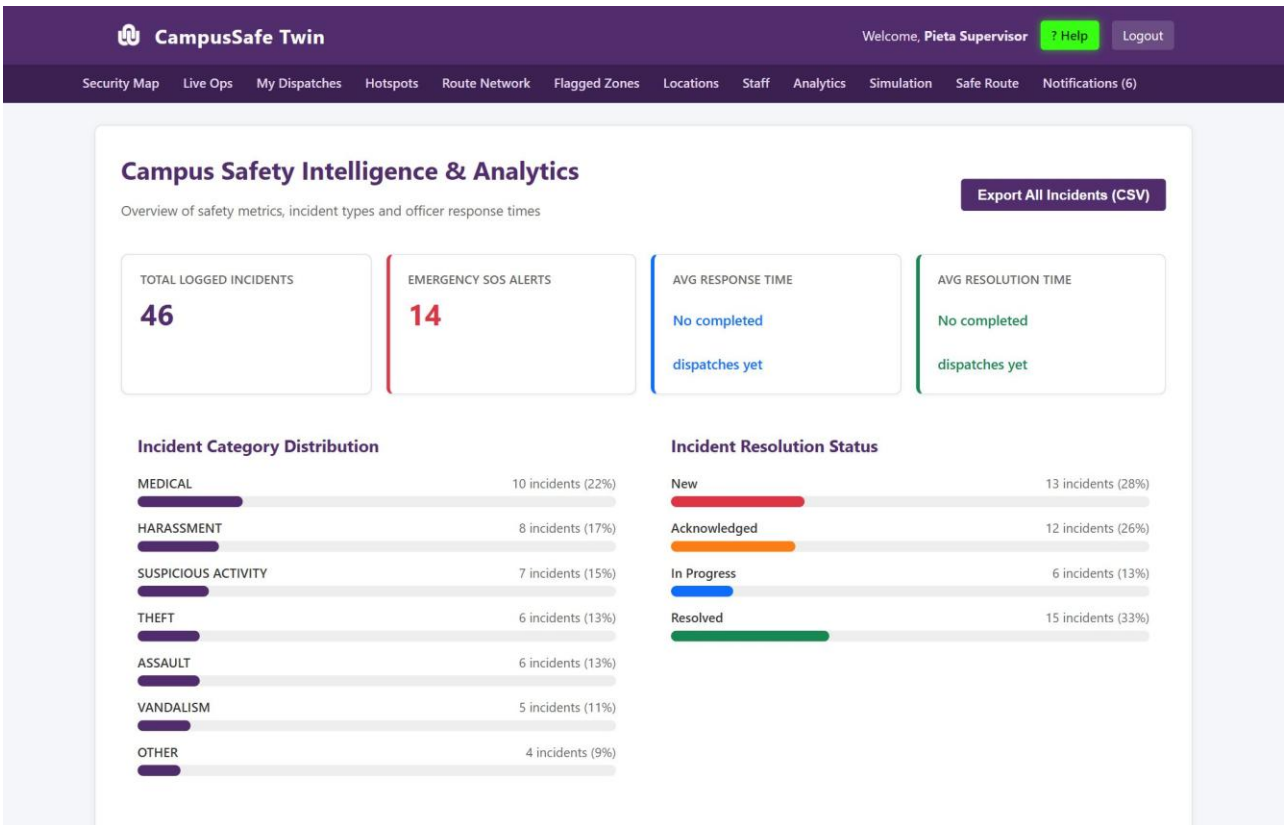


Figure 31 – After fix OBS-04: Analytics KPI tiles explaining that no completed dispatches exist yet (compare [Figure 25](#)).

8. Acceptance & Sign-off

8.1 Acceptance Statement

UAT Round 1 executed **25 of 25** planned test cases across all four modules with a **100% pass rate**. The exit criteria in [Section 3.3](#) are met: no Severity 1 or 2 defects are open, and the four low-severity observations logged during execution ([Section 7](#)) have all been **fixed and re verified in the live UAT environment**, with before/after evidence in Sections 6.1 and 7.1. On the basis of the evidence in this document, the CampusSafe Twin system is recommended for business acceptance **without conditions**.

Note: the fixes for OBS-01 to OBS-04 were verified individually against the affected screens ([Section 7.1](#)). No regression of unrelated functionality is expected from these changes (two are stylesheet/markup-only, one caps a map zoom level, one changes display text); full re-execution of Round 1 was therefore not required.

8.2 Decision

	Decision (tick one)
☐	Accepted: the system is approved for operational use as tested.
☐	Accepted with conditions: approved subject to the remediation of the observations listed in Section 7 .
☐	Rejected: material defects prevent acceptance (attach rationale).

End of document. CampusSafe Twin UAT v1.1, generated 03 October 2026.